

Choose your own coping adventure WIP title

**A THESIS
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There are many people that have earned my gratitude for their contribution to my time in graduate school.

Dedication

To those who held me up over the years

Abstract

This thesis presents *The Archive*, a narrative-based mobile serious game designed to support stress management among college students. Using a Research through Design (RtD) framework across four phases — a survey of student stress and gamification preferences, participatory focus groups, a five-day deployment with 18 participants, and an expert evaluation — we investigated how gamification mechanics can be reconsidered through co-design to better support student wellbeing. Quantitative findings indicate modest improvements in perceived stress and flourishing over the deployment period. Post-survey correlations suggest that narrative agency, emotional authenticity, and eudaimonic engagement are associated with positive outcomes. Expert evaluation identified narrative pacing and non-punitive engagement as particular strengths. We discuss implications for the design of eudaimonic games and participatory approaches to college mental health technology.

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Chapter 1

Introduction

Those who choose to attend college range widely in identification, ideology, ethnicity, and even age. Secondary education welcomes all who dare to enter its halls and endure the trials deemed necessary to succeed. This is an academic standard to which all are subjected, regardless of where they hail from or how arduous it was to arrive.

In 2024, the American College Health Association reported that 76.4% of undergraduate students surveyed experienced moderate or high levels of perceived stress within the previous 30 days (American College Health Association, 2024). At a University level, we see a similar pattern take shape. The Boynton Health 2024 College Student Health Survey conducted at the University of Minnesota Duluth found that 44.5% of surveyed college students were unable to manage their stress. Additionally, this report identifies stress as the top personal issue affecting these students, reporting that it impacted their academic performance (University of Minnesota, Boynton Health, 2024).

Alongside the current stress data from college students, counseling centers report rising clinical needs. In 2006, a survey of 274 counseling center directors revealed that 88% reported an increase in severe psychological problems over the prior five years (Gallagher, 2006). More recent work evaluates college student service utilization over a 10-year period, citing an increase in services (Lipson, Lattie, and Eisenberg, 2019). This leads some practitioners to look for ways to extend their care services in accessible ways that can meet the needs of college students. Traditional college students are classified as young people between the ages of 18 and 22 who are just leaving high school and starting a more independent lifestyle away from family members' support (Pedrelli et

al., 2015). This transition from living under a form of security into independence can leave some with a piqued sense of distress (Leppink et al., 2016). The move into higher education typically involves managing academic loads, forming new relationships, and assuming adult responsibilities, often before students have developed the coping skills those demands require (Misra and McKean, 2000; Worsley, Harrison, and Corcoran, 2021). Prior work observes that by the mid-twenties, roughly three-quarters of those who will have a mental health disorder will have had their first onset (Kessler et al., 2007). Barriers to access in care can shape when and whether a student seeks help (Auerbach et al., 2018; Duffy, 2023). Although college campuses often have diverse student bodies, this work evaluates stress among traditional college students, as the sample studied fits the traditional definition.

Stress for many is rarely tied to a single cause and is often one rooted in personal meaning based on various factors (Deasy et al., 2014; Karyotaki et al., 2020). It is contextual, an emergent experience shaped by individual history and environmental circumstances. Students appraise and reappraise the meaning they assign to stressful events (Lazarus and Folkman, 1984). A missed deadline, for example, may be a manageable inconvenience for one student but a crisis for another whose scholarship or employment depends on it.

Universities often recognize this diverse need and offer support services such as wellness screenings, peer-to-peer programs, counseling, and mental health education (Eisenberg et al., 2013; Lipson, Lattie, and Eisenberg, 2019). Even so, college students are still reluctant to seek care, often not engaging in help-seeking behaviors until conditions worsen or impact daily living (Kustimah, Kumalasari, and Meindy, 2023). Literature suggests that college students primarily manage their acute to moderate stressors informally with self-help strategies or social support (Ebert et al., 2019; Hunt and Eisenberg, 2010). This preference for an informal approach instead of the diverse offerings each university has may be due to time limitations or a lack of perceived need (Ebert et al., 2019; Weissinger et al., 2024). Even though students may struggle with help-seeking behaviors or time, mental health service utilization on US college campuses has increased, and the stigma around seeking help has decreased (Lipson, Lattie, and Eisenberg, 2019).

These considerations, including challenges in determining when to seek assistance

and external obligations, present an opportunity for universities to explore digital resources. A study by McCarthy et al. explores this space with first-year college students evaluating their attitudes toward mobile interventions, highlighting that stigma and embarrassment often deter students from seeking professional resources (McCarthy and Horwitz, 2025).

Digital mental health (DMH) apps, with their portable architecture, aim to support well-being by providing features such as education or therapeutic skills practice (Torous et al., 2018). Work in identifying how college students feel about these technologies illustrates that they are described as convenient and accessible relative to in-person care (Lattie, Lipson, and Eisenberg, 2019; McCarthy and Horwitz, 2025). Digital alternatives can provide anonymity and flexibility for a group with limited time or potential embarrassment about seeking help (Auerbach et al., 2018; Ebert et al., 2019). Work in HCI has begun to treat these DMH apps as a design problem, in which prototypes and artifacts are created with the target demographic to support a need. Lattie et al. argue that DMH tools should be designed with the social and contextual needs of this group in mind, since how tools are structured determines not only whether students engage but also how meaningfully they do so (Lattie, Kornfield, et al., 2020; Lattie, Lipson, and Eisenberg, 2019).

Research into DMH applications typically personalizes solutions to meet the need of a condition in tandem with therapeutic modules or exercises. A study by Livermon et al. used Cognitive Bias Modification to alter thought patterns and increase flexibility to reduce rigid negative thought patterns (Livermon et al., 2025). Although the results did appear to help some of the participants in this work, it was noted that further personalization should be done to enhance the users' ability to engage effectively. Work in design research by Lattie et al. with college students on their digital mental health preferences revealed these tools should consider options such as providing isolation, peer support, sensitivity about clinical language, and availability of resources that can be applied to their lives (Lattie, Kornfield, et al., 2020).

DMH apps that are currently on the market do not offer some of the features identified in prior studies (Cheng et al., 2019; Torous et al., 2018). They instead shift their focus to broad approaches for wide audiences that may fit transdiagnostically. DMH

applications such as Headspace offer modules with exercises based on therapeutic practices, such as body scanning or controlled muscle tension, to regulate the nervous system during intense moments (Gavrilova and Zawadzki, 2023; O’Daffer et al., 2022).

Although these skills are rooted in clinical therapies, these commercial applications, like Headspace, try to shift the language to be user-friendly and include audio for guided skill building. Research on the effectiveness of these DMH apps shows mixed results in comparison to their paper-format alternative (Carla Oliveira et al., 2021). The benefit seen within the DMH applications does not rest solely on the effectiveness of the therapeutic solution but on how maintainable the solution is within a person’s daily life (Linardon and Fuller-Tyszkiewicz, 2020; Lipschitz et al., 2023).

To address this, DMH apps incorporate gamification—the use of game design elements in non-game contexts (Sebastian Deterding et al., 2011)—as a way to drive engagement and sustain user behavior in digital wellness applications (Cecilia Cheng and Omid V. Ebrahimi, 2023; Six et al., 2021). While the approach has intuitive appeal, many such tools struggle with high attrition and poor sustained engagement (Linardon and Fuller-Tyszkiewicz, 2020; Lipschitz et al., 2023; Litvin et al., 2020). A common misconception is that adding game elements alone produces a meaningful experience. Dah et al. calls this shallow gamification: surface-level use of game elements that leaves the underlying experience untransformed (Dah et al., 2024).

However, it appears that this shallow gamification is a common occurrence within DMH apps on the market. A systematic review of gamification in mental health and well-being applications found that the most commonly utilized mechanics include points, levels, rewards, and progress feedback (Cheng et al., 2019). The benefits of these components are their simplicity to implement into a system and the straightforward visualization of user progression and stat tracking. However, some works suggest these reward-driven mechanics rarely produce durable intrinsic motivation (Mekler, Brühlmann, et al., 2017; Nicholson, 2015). They target hedonic value, emphasizing components that provide external rewards while being unable to provide long-term internal rewards. Research has begun to shift this one-sided balance with eudaimonic engagement (Iacovides and Mekler, 2019; Mirhadi, Iacovides, and Denisova, 2024). Rather than focusing on stat tracking or badges, eudaimonic components support meaning-making, reflection, and growth, which are more closely tied to mental wellness. This re-balancing away

from hedonic-focused reward systems within research affects the way in which DMH applications are created for the general population, but for college students, design research suggests finding this balance may be one sought alongside the students themselves (Lattie, Kornfield, et al., 2020; McCarthy and Horwitz, 2025).

Researchers have turned to participatory methods, which engage users as collaborators in design through mutual learning and collective action (Bjögvinsson, Ehn, and Hillgren, 2012; Muller and Kuhn, 1993; Sanders and Stappers, 2008; Spinuzzi, 2005). Partnering with the student community presents an opportunity to identify feature requirements or surface how stressors shape academic life.

In work by Na et al., co-design sessions with college students created a prototype Mendorphins that allowed for sharing experiences of stress and planning self-care strategies (Na et al., 2022). This pilot study showed promising results for students, allowing some symptoms to decrease. This work suggests that DMH applications may see benefit when they align closer with stakeholders and practitioners. The co-design process may have been what helped strengthen this solution (Lattie, Kornfield, et al., 2020; Severes, Barreto, and Esteves, 2025). Using these design methods has been a popular way in HCI literature for gathering qualitative insights from groups rooted in meaning through co-creation strategies (Halverson et al., 2018; Muller, Wildman, and White, 1994). With the DMH design process in mind, students are best positioned to articulate how their stress is shaped and what supports would feel meaningful to them.

But how does one create a DMH application that incorporates components such as meaning-making, reflection, and growth? A small but growing body of work suggests that narrative-driven design may support meaningful engagement with mental health content (Desai et al., 2021; Kitromili, Hargood, and Skains, 2025). Narrative and story-driven mechanics offer a focused, creative pathway to stress relief, supporting identity exploration through worldbuilding and enabling emotion- and meaning-focused coping (Iacovides and Mekler, 2019; Kim, Moon, and Chung, 2025; Mirhadi, Iacovides, and Denisova, 2024). Kitromili et al. illustrate this in their study of *Betwixt*, where participants demonstrated a preference for the interactive storytelling and the sense of agency it provided (Kitromili, Hargood, and Skains, 2025). However, *Betwixt* was developed with writers and therapists rather than through participatory design with college students, so we read this as a piece of Digital Interactive Narrative Fiction where

the expertise of professionals shaped the product. On the other hand, research-oriented design has created applications such as COEX-IST that move closer to community-informed design. COEX-IST builds story modules from the personal experiences of undergraduate students to make depression and isolation more recognizable in narrative form (Rodrigues et al., 2022).

Although these tools address general mental health themes, they do not always capture how college students navigate stress. To help support capturing these experiences, appropriate data-gathering methods leveraging demographic insights need to be employed. To support a DMH gamified tool that college students find enjoyable, we look to design methods that prioritize the stakeholders. The open design problem, then, is how to build narrative wellness tools with college students rather than for them, treating students’ lived experience as the primary source of design knowledge.

To address this problem, we adopt Research through Design (RtD) as the methodological backbone of this thesis. Frayling distinguished research into, through, and for design, arguing that knowledge can be produced through the process of making rather than only through observation or testing (Frayling, 1994). HCI later operationalized this through the RtD paradigm, which treats iteration and artifact-making as legitimate research practices (Fallman, 2005; Zimmerman, Forlizzi, and Evenson, 2007).

When applied to wellness tools for college students, meaningful design knowledge about what supports students, what engages them, and what they reject emerges through the process of making and iterating with them. Within this framework, we use participatory design with playful probes as our primary method of inquiry. Playful probing expands cultural probes by developing a game tailored to the topic under study, which is found to deepen participant engagement and produce richer data than standard probe materials (Bernhaupt et al., 2007).

Stress can be sensitive to discuss directly, and conventional interviews can elicit social desirability or self-protection. However, tabletop games give students a fictional frame in which they can voice stress through characters and scenarios. This can allow for the surfacing of experiences that they might not share when asked plainly. Role-play has growing documented use in mental health contexts (Arenas, Viduani, and Araujo, 2022; Baker, Turner, and Kotera, 2022), and using playful inquiry to study narrative

wellness aligns with the artifact we are designing. We pursued this work through a three-phase study with college students at a single mid-sized U.S. university, culminating in a five-day field deployment of the resulting system with 20 students.

The process culminated in the design and deployment of *The Archive*, a narrative-centered mobile system for college student stress. Rather than relying on reward-driven mechanics common in wellness gamification, the design rests on a set of principles that emerged from the synthesis of our three-phase co-design. These principles were derived from the Phase II focus group data, and Chapter ?? develops them in detail.

The aim of this work is: to invite students to choose their own coping adventure rather than follow a prescribed wellness pathway and to evaluate how co-designed DMHs with story features support stress relief. As story-driven narratives in games—and gamified applications—aim to prompt personal meaning in the fiction, this work aims to investigate the design process of gamified DMH applications. This work orients towards eudaimonic engagement: meaning-making, reflection, and authentic self-expression rather than habit formation through repeated hedonic rewards.

This work responds to a feature of the appraisal model: students decide what their stress means and what response feels meaningful to them. *The Archive* is not a clinical intervention or a replacement for therapeutic care. It is a research-through-design artifact that gives us a concrete site for studying what narrative-centered tools may need to offer to be meaningful for this population. We describe the system in detail in Chapter ??.

- **RQ1:** How does engagement with a narrative-centered wellness application affect students' perceived stress?
- **RQ2:** Which game-based design elements shape students' engagement with stress-related stories, including their sense of agency, narrative connection, and emotional response?
- **RQ3:** What design decisions emerge from co-designing a narrative-centered wellness application with college students?

This thesis offers five contributions, framed within the interpretive scope of an exploratory RtD study. First, it offers an empirical account of how college students in

our focus groups articulated academic stress through narrative metaphor and collaborative world-building, surfacing recurring spatial framings of stress as something inhabitable rather than something measurable. Second, it presents a participant-grounded critique of common wellness application design patterns, identifying where compliance-oriented mechanics may misalign with how students in this study described engaging with stress. Third, it derives a set of three design principles for narrative wellness systems and explores them through deployment rather than offering them as validated design rules. Fourth, it contributes *The Archive* itself as a research-through-design artifact, instantiating these principles in a working mobile system and reporting preliminary evidence on perceived stress, flourishing, and game-experience quality from pre- and post-deployment measures. Fifth, it contributes methodological resources for future work in this space: a customized tabletop role-play probe that surfaced stress narratives without requiring direct self-disclosure and an in-application ecological momentary assessment design that captured in-the-moment experience alongside retrospective measures.

Chapter 2

Background

This chapter explores topics relevant to the design of an artifact co-created with college students to identify ways to support identified academic stressors. Because the sample consisted predominantly of traditional college students, the chapter begins by examining how this population experiences stress and what support methods they employ today. Due to this demographic's relationship with technology, the chapter also examines how designers, alongside college students, can work together to share meaningful experiences that shape gamified DMHs. This section provides context for the preferences from our sample showing trends across interdisciplinary spaces. The chapter covers the blending of these disciplines: co-design, stress management strategies, digital mental health (DMH) tools, games and gamification, the mechanics that make those games, fiction and ergodic literature, and the balance of ludic play. Together, these topics provide context to the decisions made and how the preferences of our sample align with the trajectory of these many disciplines.

2.1 Stress, Appraisal, and the Limits of Standardized Tools

Stressors arise not just from events or conditions, but also from how a person feels about their environment. The transactional model of stress, presented by Lazarus and Folkman, posits that an event becomes a stressor only when it is appraised as threatening within the limits of the environment (Lazarus and Folkman, 1984). Each event, therefore, produces a different outcome based on resources and interpretations. This model

demonstrates how stress is navigated psychologically using an appraisal and reappraisal method. In practice, if a student fails an exam, they will emotionally and intellectually come to terms with their failure. This is appraising the stressor. From there a plan of action is formulated, taking into account deadlines and personal resources. This is the process of reappraising the stressor, and it is initiated so the student can reevaluate and implement the appropriate coping style. It is important to note that anything that a person does to deal with stress is defined as a coping mechanism. So tackling the stressor head on, and therefore taking actions to change the situation entirely, is considered an action to cope with stress. However, not every coping response is considered healthy or productive. Some are dissociative and promote avoidant behaviors. Those with maladaptive coping mechanisms may not have the emotional stability to navigate the stressor. This pattern appears in students who assess a conversation with a peer or instructor as a stressor. To them it may become more burdensome to reach out than to struggle in silence.

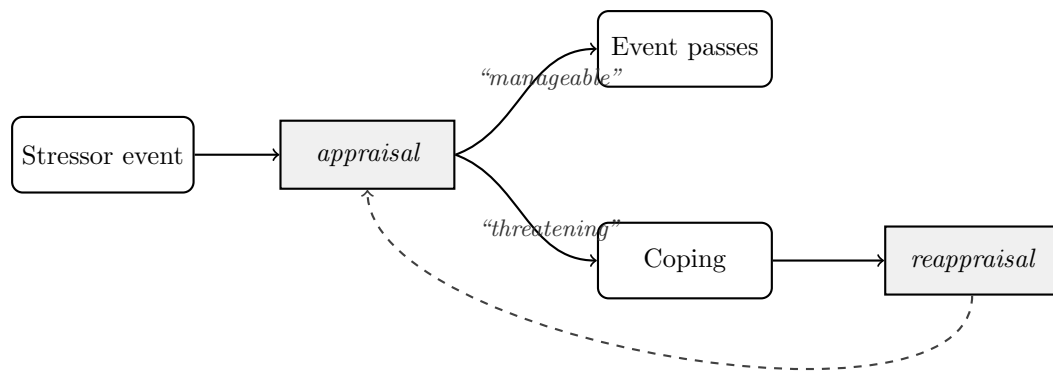


Figure 2.1: Stress is interpreted, not received (Lazarus and Folkman, 1984). The student’s appraisal is the fork: the same event passes through one student and lodges in another. Reappraisal returns the student to the fork as the situation unfolds.

Coping is defined as the cognitive and emotional strategies used to manage events perceived to be challenging (Lazarus and Folkman, 1984). This process relies on adapting behaviors appropriate for the event. Prior works ground these adaptive coping behaviors in two orientations: problem-based behaviors that tackle the event directly

and emotion-based behaviors that regulate emotional reactions around the event. The coping style used by students directly influences the way perceived stress affects mental well-being (Slimmen et al., 2022). Thus, beginning to understand the meaning of stress from the perspective of students and how they apply coping skills is vital for designing DMH applications that support specific socio-demographics. Work by Park and Folkman extends the stress model by describing meaning-making in the appraisal process (Park and Folkman, 1997). Global meaning refers to factors that align more intrinsically, shaping a person's fundamental assumptions about the world. Various aspects can shift these influences, such as region, economic status, or other life experiences. Situational meaning aligns more closely with how academic stressors are appraised. This framework has three major components: assessing personal significance, the coping process if the situation is stressful, and the aftermath. These three components are evaluated against the individual's global meaning and the overall goals of the event. The framework of meaning from the appraisal process suggests that students must be continuously prioritizing their academic goals, so our work emphasizes this perspective of stressful experiences in developing support.

In work by Slimmen et al. a cross-sectional survey was used to identify underlying academic stressors of college students in Applied Sciences (Slimmen et al., 2022). Academic pressure was found to be the top identified stressor, consisting of items such as pressure to perform, perceptions of workload, examination, self-perception, and time constraints. Each student may react differently to these underlying events, but many of these stressors present themselves as components of academic stress. In therapeutic work, maladaptive responses to these events are supported by interventions to change behavior patterns or cognitive restructuring. Therapeutic frameworks such as Cognitive Behavioral Therapy (CBT) provide structured talk therapy that many therapists use to support symptoms of distress (Hofmann et al., 2012). Due to its popularity and flexibility to support transdiagnostic symptoms, these techniques have been used as a support across different modalities (Hofmann et al., 2012). CBT-aligned interventions have been shown to be effective when adhered to, and therapeutic guidelines suggest that new patients should complete six or more face-to-face sessions clinically (Robinson, Kellett, and Delgadillo, 2020). For college students, some of the stressors identified may be barriers to adherence, which is where these tools shift modality to allow agency in

care.

Commercial DMH applications aim to increase flexibility in wellness support by allowing users to access interventions as self-guided routines. Apps such as *Headspace*, *Finch*, and *Betwixt* are widely available through the *Google Play Store* or the *Apple Store*, where users simply download the software onto their mobile device. Each is advertised to help with stress management and incorporates many of the treatment goals CBT contains. The principles of this structure connect thoughts, feelings, and behaviors (National Institute of Mental Health, 2026). One of the key aspects of CBT is learning to be your own therapist through experimentation. This allows patients to focus on building their problem-solving skills to test out possible fears related to stressful events. Each of these applications leverages this approach in a different way, allowing users to build resilience to stressful events supported by the mobile architecture to shift self-identified maladaptive behaviors.

For example, *Headspace* cites a CBT program that is rooted in this therapy, allowing users to understand the relationship between their thoughts, feelings, and actions (Headspace, 2024). The experiential nature of this process allows a variety of different methods to take root. *Headspace* allows the user to self-reflect on behaviors to explore perspectives and reframe negative thought patterns. This model uses a unique approach to allow its users to understand their thought patterns and identify their behaviors. Through video modules, stories of individual struggles with mental health are portrayed to give the audience a relatable starting point. This allows watchers to start with establishing similar goals to those within the introduction sequence, aligning their vision of better well-being with how it is represented by others. Each following module is unlocked one activity at a time, featuring meditation exercises or thought-reframing prompts. *Headspace* blends short videos to provide education on mental health along with flexible exercises. The sequence followed in this structure models many of the tenets of CBT therapies. One of the downsides of this model is that it relies on self-disclosure or having awareness of one's current emotional needs, which may be difficult for some developing college students. Additionally, models such as these are cited to be difficult for college students to engage with due to the overtly clinical language, consistent time commitment, and lack of representation of embodied stress for this group (Lipschitz et al., 2023; McCarthy and Horwitz, 2025). Some DMH apps shift to provide

more specific support by turning self-care tasks, which are often left unmet during times of struggle, into a visualized game with achievable goals.

The DMH application *Finch*, for instance, provides support in a less clinical way by prioritizing the completion of self-care. For those whose stressors are impacting daily living, this structure provides support by validating small and manageable goals to support habit formation. Due to the nature of the identified stressors in academia, students can miss self-care activities, such as healthy sleep routines or exercise. Work by Moses et al. examined the role of self-care practices among college students, finding a significant relationship between psychological adjustment and self-care (Moses, Bradley, and O’Callaghan, 2016). For a population that is still developing the cognitive skills to navigate adulthood, maintaining these self-care activities does not always come naturally and may lead to negative mental health problems if not prioritized over time. These care habits include items like mindfulness, going for a walk, and taking a deep breath, which can boost mood by lowering stress and improving well-being. When basic care tasks become a struggle for students, betterment must begin with personal initiative. Awareness of self-care must come first, often through demonstration or encouragement. Finch models the encouragement aspect of this recommendation. This application partners goal setting with a friendly virtual pet companion that celebrates goal achievement, making a game out of meeting self-care needs. If the user’s goals align with the application, and they have awareness of their own behavior patterns, models like these can support existing structures. Limitations appear with the type of gameplay provided and restrictions aligning with the type of play presented.



Figure 2.2: Finch’s pet-caretaker mechanic. The daily task list (left) lets the user log self-care actions, which earn points spent on customizations for a virtual companion (right).

When these applications are evaluated on their efficacy, results are found to be mixed. A randomized controlled trial on the meditation features of *Headspace* over a 2-month period found anxiety symptoms improved but no significant improvement on worry (D. Abbott, Lack, and Anderson, 2023). Although the study noted that the control condition involved a more intensive therapeutic intervention, the fact that a low-contact method produced similar results was impactful. This finding suggests that digital methods may provide benefits to users who do not engage in face-to-face treatment. The flexibility is especially relevant for college students whose stressors revolve around shifting time commitments. Their value comes from the portability: they remain accessible digitally for students whose schedules are limited or shifting.

2.2 Narrative as a Coping Modality

Fiction portrays imaginary individuals, events, or places, and audiences approach these works expecting deviation from the real world rather than factual representation. The themes within a fictional work are open to interpretation, and how those themes relate to lived experience is left for the audience to assign meaning to. This interpretive property is what therapeutic practices and game design have drawn on to support emotional processing.

Narrative therapy applies this interpretive frame in a clinical setting. The therapist works with the patient to externalize the problem, identify episodes that contradict the dominant story, and reauthor the narrative around those episodes rather than around the original framing (Vromans and Schweitzer, 2011). The mechanism is interpretive rather than behavioral. The patient is not asked to change what happened. Instead, they are asked to change the context the event sits within. Once recontextualized, the meaning of the story can change, and that determines whether the event continues to act as a stressor. Vromans and Schweitzer evaluated this mechanism in a one-sample trial of eight-session manualized narrative therapy with 47 adults diagnosed with major depressive disorder. Seventy-four percent of participants achieved reliable improvement in depressive symptoms, a large effect size comparable to benchmark outcomes for established psychotherapies (Vromans and Schweitzer, 2011).

The same authorial structure that makes narrative therapy work also lets story-based games carry similar weight. In both, the participant has control over how an experience is portrayed and resolved. Tabletop role-playing games have been studied in this register. Abbott and colleagues, reviewing tabletop role-playing games as a therapeutic intervention with adults, report consistent gains in social connectedness and personal growth attributed to the narrative transference and collaborative authorship the format enables (M. S. Abbott, Stauss, and Burnett, 2022). Peralta extends this evidence into a college context. In a quasi-experimental study with first-semester students, eighteen participants who attended five weekly Dungeons & Dragons sessions showed a significantly greater reduction in college stress and stronger personal-emotional adjustment than a comparison group (Peralta, 2024). The transfer runs back into lived experience: the player practices a version of themselves within the fictional frame and carries that

practice into the academic and social settings the fiction sits one step removed from.

Single-player digital narratives translate the authorial qualities of tabletop play into a different form. Rather than collaborative storytelling at a table, the player traverses a structured text in which choices change what unfolds. Aarseth describes this form as ergodic literature, where nontrivial effort is required to traverse the text and where the path taken is never fully visible in advance (Aarseth, 1997). The reader becomes an active participant whose choices shape the encounter rather than a passive observer moving through a fixed sequence. *COEX-IST* illustrates this form for mental health content. The player inhabits a protagonist followed by a materialized form of depression, and in-game decisions change how the environment and soundtrack respond as well as which ending the player reaches (Rodrigues et al., 2022). The participant is not receiving a story; they are constructing a path through it, and the construction itself is what produces engagement with the material.

For someone working through stress or emotional difficulty, this structure matters because the uncertainty of not knowing where the next passage leads, paired with the sense that choices carry consequence within the world of the story, produces a kind of immersion that linear text rarely reaches. The fictional frame provides distance from lived experience, which lowers the stakes of engaging with material that might otherwise feel too close. The authorial structure provides leverage: the participant works on the meaning of the experience by working on the meaning of the story. This is the structural alignment between narrative therapy, tabletop role-play, and digital interactive narrative. It is the alignment a narrative-centered wellness application can draw on if the design treats authorship as the primary mechanic rather than as decoration around a fixed protocol.

2.3 Gamification: Mechanisms, Player Experience, and Eudaimonic Engagement

Game elements layered onto wellness applications are routinely chosen on the assumption that adding features will increase motivation. The assumption is intuitive: attrition is high, gameful structures are popular, and an interface that feels like a game ought to keep users coming back. Empirical evaluations of gamified mental health applications

have not borne this out cleanly. A meta-analysis of gamified depression apps found inconsistent effects on engagement and outcomes (Six et al., 2021), and broader reviews ask why gamification often fails to produce the motivation it is designed to generate (Dah et al., 2024). The problem is not gamification itself but the assumption that any game feature, placed in any context, will produce a particular experience for the user. Effective gamified design rests on understanding what kind of experience a given mechanic actually produces, which requires a closer look at what play is and how it relates to the games that contain it.

The distinction between play and games is older than gamification. Caillois categorized human activities into forms of play and treated play as integral to routine human life rather than as a category opposed to it (Caillois and Barash, 2001). Huizinga, by contrast, argued that play is defined by the experience it affords rather than by its surface form: an activity is play because of the kind of attention and freedom it elicits, not because it is wrapped in the trappings of a game (Huizinga, 1955). The two views together establish a distinction that gamification scholarship continues to draw on. Deterding and colleagues formalized the distinction for design, defining gamification as the use of game design elements in non-game contexts and separating gamefulness from full games (Sebastian Deterding et al., 2011). The lesson for wellness design is that adding a badge or a streak does not by itself produce play; it produces a game element, and whether that element produces play depends on the experience it sets up for the user.

The MDA framework, formulated by Hunicke, LeBlanc, and Zubek to bridge game design and game research, gives an account of why this gap between elements and experience exists (Hunicke, LeBlanc, and Zubek, 2004). MDA decomposes a game into three layers. Mechanics are the structural rules and components of the system. Dynamics are the behaviors that emerge as users interact with those mechanics. Aesthetics are the experiential qualities the user actually feels. Mechanics are designed; aesthetics are experienced. The same mechanic placed in two different contexts can produce two different aesthetics depending on the dynamics it sets up. A leaderboard placed in a fitness application produces dynamics of competition and an aesthetic of pressure. A leaderboard placed in a wellness application aimed at college students with academic stress produces the same dynamics but an aesthetic of additional burden. MDA is therefore useful as a diagnostic for understanding why some game-based wellness designs land

while others miss. The mechanic is not the question. The question is what dynamic the mechanic sets up in the population the system is designed for.

Several commercial applications use narrative mechanics in ways that align with the desires college students describe in co-design. *Betwixt*, a mobile application designed to support mental resilience, draws on the Choose Your Own Adventure narrative form and casts the user as the protagonist of a self-discovery quest. Kitromili and colleagues studied the application in a four-week user-experience evaluation. They found that the interactive element was the main attraction for participants regardless of gaming background, and that the conversational tone and high level of user agency supported emotional engagement (Kitromili, Hargood, and Skains, 2025). The Choose Your Own Adventure structure gives the user authorship over how the story unfolds, which produces dynamics of curiosity and reflection rather than dynamics of performance. For college students who describe wanting agency without clinical framing, this structure does both at once. It offers a meaningful interaction that does not feel like therapy, while still producing the interpretive engagement that supports emotional reflection.

Other applications use immersion and aesthetic framing to lower the social and emotional cost of self-care. *Kinder World* presents a cozy indoor space in which the user practices self-care activities such as journaling, breathing exercises, and tending plants, framing the experience as inhabiting a quiet domestic world rather than completing wellness tasks. *Voidpet Garden* uses the pet-caretaker relationship to support mood tracking but anchors that relationship within an aesthetically rich garden that reframes daily emotional check-ins as care for a small living world. Both designs prioritize emotional safety, low-stakes engagement, and accessible language over clinical instruction. They address themes that students raise in co-design about wanting wellness support that does not feel like therapy. They also function as forms of escapism, which Section 2.1 identified as one of the most common coping practices among college students. They convert escapism into an opportunity to practice self-care through immersion, rather than into avoidance through distraction. In MDA terms, the mechanics are comparable to those in reward-driven applications, but the dynamics produced by the narrative atmosphere shift the aesthetics from performance pressure toward gentle engagement.

Beyond purpose-built wellness applications, commercial narrative games have been

shown to support adaptive coping during difficult life experiences. Mirhadi and colleagues’ exploratory survey of 130 players found that games supported emotion-focused, avoidance, and meaning-focused coping during periods of personal difficulty, with specific game aspects mapping onto specific coping strategies (Mirhadi, Iacovides, and Denisova, 2024). *Lifeline*, a text-based interactive fiction in which the player makes decisions that affect a stranded astronaut’s survival, places the user in a relationship with a character whose emotional state mirrors and extends the player’s own. *Depression Quest* depicts the texture of living with depression through hyperlinked choices, some of which are visibly removed from the player’s reach as the story progresses, mirroring the way depression constrains agency in lived experience. *Underworld Office*, a mystery game set in a supernatural workplace, pairs lighter aesthetics with branching choices about social and ethical pressure. None of these are clinical interventions, and none use clinical language. Their value as adaptive coping support comes from the way narrative immersion lets a player relate to a story whose structure mirrors a difficult experience, an effect that has been observed to support cognitive functioning during stressful periods (Desai et al., 2021; Iacovides and Mekler, 2019).

Self-Determination Theory (SDT) provides a theoretical frame for why these designs produce engagement where reward-driven gamification does not. SDT identifies autonomy, competence, and relatedness as basic psychological needs whose satisfaction supports intrinsic motivation and well-being (Ryan and Deci, 2000). The narrative-centered designs above support all three. The user authors the path through the story, which supports autonomy. The user navigates the story successfully, which supports competence. The characters and narrative situations mirror the user’s emotional state, which supports relatedness. Tyack and Mekler caution against superficial applications of SDT in HCI and serious-games research, and the distinction between superficial and substantive autonomy holds here (Tyack and Mekler, 2024). A wellness application that frames autonomy as a choice of when to log into the product is structurally different from one that frames autonomy as authorship over how the experience unfolds. The first is autonomy of access; the second is autonomy of meaning.

The deeper distinction underlying these design contrasts is between hedonic and eudaimonic engagement. Hedonic engagement is oriented toward immediate pleasure,

reward, and entertainment. Eudaimonic engagement is oriented toward meaning, personal growth, and the development of one’s potential (Huta and Ryan, 2010). Mekler and Hornbæk extend this distinction into HCI, finding that eudaimonically motivated experiences and social experiences are perceived as more meaningful than hedonic experiences focused on relaxation or short-term pleasure (Mekler and Hornbæk, 2019). Reward-driven gamification mechanics target hedonic value almost exclusively. Narrative mechanics, by contrast, can produce dynamics such as curiosity, reflection, and interpretation that align with eudaimonic engagement, because the player is asked to construct meaning rather than to perform behavior. A wellness system can move in this direction without abandoning gameful elements. It can replace ranking with shared progress, replace streaks with narrative continuity that the user controls the pacing of, and replace badges with story consequences that follow from the player’s choices. Each of these moves keeps the underlying gameful structure but changes the dynamics the mechanics set up, and therefore the aesthetics the user experiences.

The contrast between hedonic and eudaimonic engagement is the alignment a narrative-centered wellness application can draw on. Reward-driven gamification reaches users through the surface of the game; narrative-centered design reaches them through authorship of meaning. For a population whose stressors revolve around shifting time commitments and an academic environment that already rewards performance, the latter offers something the former cannot. It treats engagement not as compliance with the system but as a form of working through the experience the system is designed to support.

2.4 Research through Design and Participatory Inquiry

Measurement tells us whether a system produced its intended experience. Participatory work tells us whether the intended experience was the right one to produce in the first place. The case for participatory grounding in digital mental health design is sometimes treated as self-evident. It is not. We advance participatory design as a constitutive method rather than a supplementary one in this thesis, which means defending that choice against alternatives that have produced engagement and outcome metrics without it. The methodological frame for that defense is Research through Design

(Frayling, 1994). RtD treats artifact-making and iteration as practices that produce design knowledge of their own kind, rather than as steps preceding evaluation (Fallman, 2005; Zimmerman, Forlizzi, and Evenson, 2007).

What outcome-driven evaluation cannot recover is the design knowledge about why some students would not self-select into a wellness study, or what design moves would have made a system resonant for them. Sections 2.1 and 2.2 surfaced themes that align with what students articulate when given voice in co-design: a preference for non-clinical framing, the value of agency without performance pressure, and the use of escapism as a vehicle for self-care rather than as avoidance. These themes are not deducible from clinical literature or usability testing. They emerge when students are given the materials to articulate what stress feels like and what kind of support would meet it. RtD provides the iterative scaffold for that articulation, and participatory inquiry provides the substance.

Participatory design has limits, and the same literatures flag them. Malloy and colleagues’ scoping review of eighteen co-design studies in youth digital health reported that user experience testing was the most common evaluation outcome across the included studies, while deployment-stage evidence appeared in only a small fraction of cases (Malloy et al., 2023). For a wellness tool meant to support stress that fluctuates across an academic semester, the gap between workshop-stage feedback and deployment behavior is the gap that determines whether the tool works when students engage with it on their own schedule. The same review observed that co-design with young people often emphasized aesthetic preferences and feature-level critique rather than the underlying assumptions of the system being designed (Malloy et al., 2023), which means a co-design that surfaces only what students like the look of can leave the deeper questions of how stress should be framed unaddressed.

These limits do not undermine the case for participatory design; they shape it. They mark the design questions this thesis takes up: how to carry participatory inquiry past the workshop stage and into deployment, and how to keep co-design oriented toward the assumptions of the system rather than its surface.

2.5 Probes, Deployment, and the Service Landscape

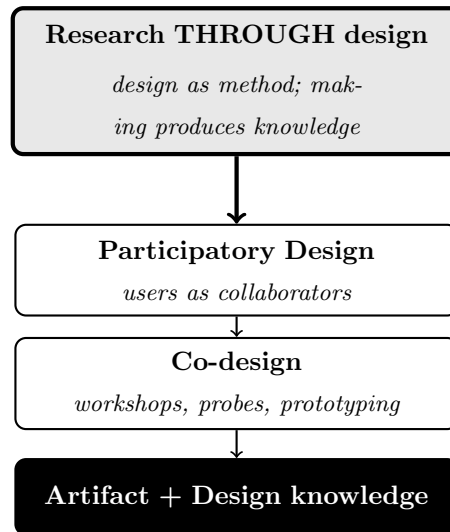


Figure 2.3: This work locates itself in Frayling’s third mode of design research, where making produces knowledge (Frayling, 1994). Participatory Design and co-design are the HCI methods that operationalize that mode for this thesis: PD frames who participates and on what terms, and co-design names the specific workshops and probes through which the artifact is made.

Design research has long faced a practical challenge: how do you access experience that people have not yet found words for? Standard interview formats and surveys depend on articulation that is already settled. Participants report what they know how to say, which tends to reproduce familiar framings of familiar problems. For experiences that are emotionally complex, contextually variable, or socially embedded, such as the stress college students navigate across a semester, these methods often reach the surface but not the depth beneath it.

Cultural probes were introduced by Gaver, Dunne, and Pacenti as one response to this challenge (Gaver, Dunne, and Pacenti, 1999). Working with elderly residents in three European communities, the researchers distributed packages containing postcards, maps, cameras, and other open-ended artifacts and asked participants to document, reflect, and respond over time. The returns surprised the team in their range and texture, surfacing routines, dispositions, and lived realities that conventional interviews had not

reached, and the resulting design proposals for community-supportive technology drew their substance from those returns rather than from a structured needs assessment. Probes are deliberately incomplete: they create the conditions for meaning to emerge rather than specifying what the meaning should be. For a population whose experience is socially embedded and varies across moments, this matters because it lets the participant bring forward what they did not know was relevant until the materials prompted them to.

Playful probes build on this foundation by introducing game elements into the probe itself. Bernhaupt and colleagues developed the playful probing method as an extension of cultural probing, designing games tailored to the domain under study and using play as the primary mode of participant engagement (Bernhaupt et al., 2007). In their study with households, playful probes produced roughly twice as much data as standard cultural probes, and the data was richer in personal detail and contextual texture. The game format lowered the social stakes of participation and gave participants a structure within which meaning could surface through action rather than through retrospective account. Loi extends this lineage with playful triggers: creative, ambiguous artifacts designed to elicit collaborative practices and surface insights that conventional processes fail to materialize (Loi, 2007). Across both lines of work, the same property holds. Play, unlike direct questioning, does not require the participant to already know what they think. It lets them find out through the process of doing.

This property matters specifically for studying college student stress and gamification preferences. Stress as an experience is contextual and fluctuating. Students do not always have clear access to why a particular moment felt unmanageable or what a wellness tool would have needed to do differently to feel supportive rather than pressuring. Placed inside a structured play scenario that mirrors the social and emotional dynamics of academic life, students reveal patterns of interpretation and coping that direct questioning rarely surfaces. The fictional frame provides the symbolic distance that makes genuine reflection possible. Within the probe, participants are not asked how they would respond to stress; they respond to it, and the researcher observes how that unfolds.

Recent work in this register shows what this looks like in practice for student stress.

Yamamoto and colleagues conducted co-design workshops with graduate students structured around play activities, beginning with the sculpting of stress monsters as a warm-up to externalize what stress felt like before any discussion of design themes (Robledo Yamamoto et al., 2023). The monsters gave participants symbolic distance from the stressors they were describing. Across the workshops, the researchers identified sources of stress and coping strategies that organized into a broader design space: stress as something to control, to disconnect from, or to normalize. The output was not a feature list. It was a set of design observations and recommendations grounded in how participants conceptualized stress through play. The methodological lesson holds for our own use of probes. Play is the mechanism through which themes surface, and themes are the substrate from which design observations follow.

The Phase II focus groups described in Section 3.3 adapts *Fiasco* (Morningstar, 2009), a tabletop role-playing game by Jason Morningstar designed for short-form scenarios driven by character relationships and unstable plans, as the playful probe through which students authored stress narratives. *Fiasco*’s structure is well-suited to this work because it requires no game master, runs in a single session, and produces escalation through character relationships rather than through external challenge. These properties made it adaptable to a stress-focused frame without losing the authorial qualities that make play probes generative.

Whether a system designed through this kind of inquiry holds up in deployment is a separate question. Three deployed gamified DMH systems represent the strongest alternative position to the argument this thesis advances. SPARX produced clinically comparable outcomes through clinician-grounded CBT design. eQuoo achieved retention gains well above its controls through general gamification principles layered with narrative content. RAIDER showed pre-post mindfulness gains through a user-centered methodology with expert consultation. Each demonstrates that meaningful engagement and outcome metrics are reachable through expert-led or user-centered processes that stop short of sustained participatory work, and each should be examined directly.

SPARX, a computerised CBT intervention for adolescent depression, was evaluated by Merry and colleagues in a multicentre non-inferiority RCT at 24 New Zealand primary care sites with 187 help-seeking adolescents (Merry et al., 2012). The intervention

produced reductions in depressive symptoms comparable to face-to-face counselling delivered as treatment as usual. Development was guided by clinicians grounded in CBT protocol, and adolescent users were involved as evaluators rather than as authors of the system’s framing.

eQuoo, evaluated by Litvin and colleagues in a large-scale RCT with 1,165 students from 180 UK universities, layered narrative content over a points-and-levels reinforcement structure and produced significant increases in resilience and decreases in anxiety and depression, with 42% better retention than the control conditions (Litvin et al., 2020). The mechanics were selected from general gamification principles rather than derived from how the target population narrates stress.

Gómez and colleagues developed RAIDER, a collaborative serious game for university student stress, using a user-centered methodology with expert consultation and validating it with eleven users who showed significant gains in learning mindfulness techniques pre-to-post (Gómez, Molina, and Collazos, 2024). Students shaped the evaluation but not the system’s framing.

These successes do not extend to the population this thesis targets. SPARX served help-seeking adolescents within a primary care system. eQuoo recruited students who chose to enroll in a wellness study. RAIDER validated with eleven users recruited as evaluators within a single university program. All three selection conditions filter for a willingness to engage with structured intervention that the broader population of stressed college students does not share. The students who manage stress informally, who do not classify their stress as severe enough to warrant formal services, and who never present to a study about wellness are absent from these samples by construction.

Beyond selection, deployment also depends on the service landscape the application enters. A wellness application for college students enters an ecosystem that already includes campus counseling centers, peer support, faculty referrals, telehealth, screening apps, crisis lines, and the informal network of friends and family that most students rely on first (Eisenberg et al., 2013; Lattie, Kornfield, et al., 2020). Service design treats this landscape as the unit of analysis, focusing on how a person experiences a coordinated set of touchpoints over time rather than on any single interaction in isolation (Stickdorn et al., 2018). For college mental health, this framing matters because the failure cases are rarely failures of a single artifact. They are failures of fit between the artifact and

the wider ecosystem the student is navigating.

The framing has limits. It is well-suited to large institutional services with stable touchpoints but less well-suited to fluctuating, individual experiences such as everyday academic stress, which is rarely the subject of a single referral or transaction. For many college students, the dominant barriers to formal services are time, access, and the belief that they do not have a problem severe enough to warrant the formal touchpoints in the first place (Weissinger et al., 2024). The student is not moving through the service landscape; they are trying to manage outside of it. Designing for this population therefore requires holding both framings: the artifact must function as a service touchpoint when the student wants formal support, and as a self-directed resource when the student does not. This dual position informed two specific design decisions. First, the application would foreground student-controlled pacing rather than a fixed clinical schedule. Second, the application would treat institutional resources as available rather than as recommended, so that engagement with the artifact does not become a substitute for, or an obstacle to, the formal services the student may eventually need.

The argument across this chapter has a single shape. Stress is interpreted, not received, which means a wellness tool that targets behavior misses the unit that determines whether stress accumulates or resolves. Narrative structures engage interpretation directly because they hand authorship to the participant. Reward-driven gamification mechanics produce behavior without engagement; narrative mechanics can produce eudaimonic engagement when designed with that aim. Participatory work surfaces the design knowledge that evaluation alone cannot recover, but only when it is deep enough, sustained enough, and connected to deployment. Playful probes provide that depth, and a service-landscape position keeps the artifact useful without crowding out the formal services students may eventually need. The methodology we describe in Chapter 3 demonstrates these commitments across our three-phase Research through Design process.

Chapter 3

Methods

This chapter describes the methodology we followed to iteratively co-design and evaluate a gamified digital mental health (DMH) application for college student stress. The methodology addresses the three research questions posed: RQ1, which evaluates the influence of narrative-centered gamified systems on student stress; RQ2, which examines how particular game mechanics shape user comfort and support; and RQ3, which examines how a codesign approach informs the development of narrative-centered systems for stress reflection.

The investigation comprised three phases of empirical work: Phase I, an exploratory survey of college student stress and preferences for gamified DMH tools ($N = 38$), detailed in Section 3.2; Phase II, qualitative focus groups using a playful probe and semi-structured discussion ($N = 30$), detailed in Section 3.3; and Phase III, a five-day field deployment and use evaluation of the resulting artifact ($N = 20$), detailed in Section 3.6. Two iterative stages connected these phases. System design (Section 3.4) translated Phase I and Phase II findings into a working artifact. Expert evaluation (Section 3.5) recruited ten faculty and staff ($N = 10$) across HCI, narrative design, and mental health to identify usability issues and inform design refinements before field deployment. The progression of these stages is shown in Figure 3.1.

3.1 Research Through Design (RtD)

Chapter 2 made the case for Research through Design (RtD) as the methodological frame this work inherits (Fallman, 2005; Frayling, 1994; Zimmerman, Forlizzi, and Evenson, 2007); this chapter puts it into practice. The artifact produced through this approach, *The Archive*, is described in Chapter 4. The application of RtD this thesis uses is co-design, following Busciantella-Ricci and Scataglini’s argument that integrating participatory research into design methodology may surface insights closed-process design often misses (Busciantella-Ricci and Scataglini, 2024; Sanders and Stappers, 2008). Stress is susceptible to interpretation in meaning, so the project required an approach that could represent participants’ lived perspective rather than only their measurable behavior. This contingency shaped the choice of a mixed-methods design that pairs validated self-report instruments with open-ended and interactional data, allowing reported outcomes to be examined alongside the accounts students give of their own engagement. The RtD paradigm offered a flexible framework to capture these types of meaningful experiences.

Co-design carries has known risks identified in prior work (Sanders and Stappers, 2008). Chapter 2 flagged the one most relevant to this thesis: co-design with young people can drift toward aesthetic preference and feature-level critique rather than engaging the underlying assumptions of the system being designed (**malloyEngagementOnlineDigital2023**). We address that through expert evaluation of the co-designed product outlined in Section 5.4. Because the thesis investigates gamification mechanics within DMH applications, we take playful design methods themselves as the entry point to co-design, on the principle that play surfaces observable dynamics a mechanic sets up; the specific probe is described in Section 3.3. The phases that produced that process are arranged as an explanatory sequential design: an initial quantitative survey establishes a baseline of student preferences across game-based mechanics, and the qualitative phases that follow interpret those patterns in terms of how participants narrate their stress (Takona, 2024). Residual limits of this arrangement are addressed in Section ??.

Gamified DMH design lacks a settled set of best practices, particularly for the population this thesis targets. Given the breadth of game features in this space and the variety of purposes they serve, the inquiry began by establishing the perceived value

of these mechanics for supporting students’ needs. The mechanics, dynamics, and aesthetics (MDA) framework (Hunicke, LeBlanc, and Zubek, 2004) provides the shared vocabulary across phases, which keeps the survey ratings, the focus-group observations, and the later system-design decisions legible on the same terms. Recent work applies the RtD–co-design combination directly in this space: Bierbooms and colleagues used a three-stage RtD process with co-design sessions to develop an educational escape room for e-mental health training, iterating from purpose and storyline through prototype to field-tested artifact (Bierbooms et al., 2025). The same logic structures this project. Each phase carries a distinct kind of knowledge that the next builds on: Phase I establishes a baseline of student preferences across the candidate mechanics (Section 3.2); Phase II examines how those preferences hold up when students enact stress through narrative play (Section 3.3); the system design and expert evaluation stages translate that knowledge into an artifact and refine it against cross-disciplinary review (Sections ?? and ??); and Phase III tests the artifact in naturalistic use (Section 3.6). Figure ?? shows the relationships between these stages.

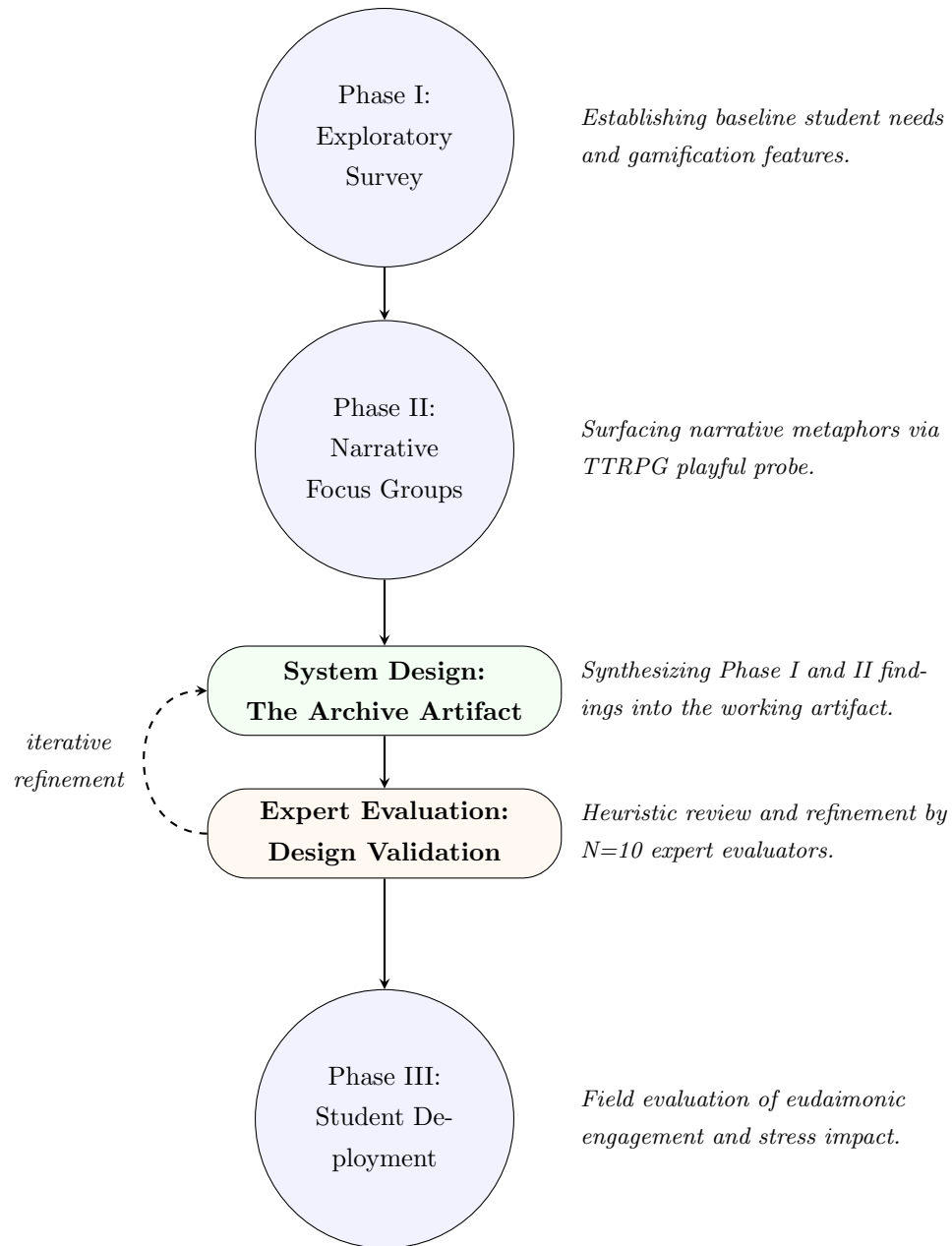


Figure 3.1: The Research through Design (RtD) process for *The Archive*, showing the progression from initial discovery (Phases I and II) through artifact synthesis and iterative expert refinement to field deployment (Phase III). The dashed loop on the left represents the iterative cycle between System Design and Expert Evaluation: review feedback shaped successive revisions of the artifact before field deployment.

3.2 Phase I: Experiential Survey

The first phase of this research used a cross-sectional online survey to examine how college students rate gamification mechanics and which self-care activities they find most useful for stress management. The survey serves RQ2 and RQ3 most directly: it produces ratings of individual game-based design elements (RQ2) and grounds the later design work in student perspectives rather than researcher assumptions (RQ3). A cross-sectional design was appropriate at this stage because the questions of interest concern students' current preferences and support needs rather than change over time, and because the outputs of this phase were used to shape a subsequent qualitative phase rather than to produce a standalone inferential claim.

The survey was administered online via Qualtrics and organized into five sections chosen to cover the decision dimensions the later design would need: *general information*, *mental well-being*, *gamification*, *social support*, and *technology and engagement*. All rating items used 5-point Likert-type scales, and several questions provided optional free-response space for participants to expand on their answers. The full instrument, with scale anchors and item wording, is reproduced in Appendix B.

The general information section collected basic demographic data, including academic major, year in college, and age. The mental well-being section identified common stressors and the coping activities students already used. The gamification section asked participants to rate specific game-like mechanics drawn from the MDA framework (Hunicke, LeBlanc, and Zubek, 2004) and Deterding's gamification taxonomy (Sebastian Deterding et al., 2011), with examples taken from gamified DMH applications and, for participants without DMH experience, from commercial video games. The mechanics included point systems and leaderboards, progress bars, personalization, story-driven quests, and collaborative activities. A follow-on subset of items in this section assessed the perceived meaningfulness of wellness-tracking features such as streak calendars and narrative journey maps, allowing us to distinguish game elements from the surrounding digital-support structure. The social support section assessed interest in group-based challenges and peer encouragement. The technology and engagement section asked participants to rate feature preferences in a hypothetical stress management application and their openness to real-time content adaptation based on sensor input.

Participants ($N = 38$) were recruited via campus flyers and in-person outreach at the University of Minnesota Duluth. Eligibility required current enrollment at the University and a minimum age of 18, and participation was unpaid. The sample skewed toward graduate students and toward majors in computer and information sciences, reflecting both the recruitment channels used and the departmental context in which the study was conducted. The sample is a convenience sample drawn from a single institution and is not intended to support population-level inference; its role in the overall design is to produce a structured baseline of student preferences that could be elaborated through the Phase II focus groups.

This phase received IRB approval prior to deployment, and the project-wide ethics protocol is described in Section ???. Findings from this phase are reported in Chapter 5 and fed directly into the design of the Phase II focus-group probe described in Section 3.3.

3.3 Phase II: Narrative Focus Groups

Phase II used the Phase I survey findings to shape the focus of the qualitative inquiry. The survey indicated a preference for narrative and exploratory mechanics over competitive ones, with story-driven quests and personalization rated highest among the five mechanics presented, and academic pressure emerging as the most strongly endorsed source of stress (see Chapter 5). Phase II serves RQ2 and RQ3 most directly. It produces interactional data about how specific mechanics function during stress-related play (RQ2), and it is the primary site at which codesign principles are enacted, with students authoring stress scenarios and critiquing existing tools rather than simply being surveyed about them (RQ3).

To examine how the Phase I preferences played out in interactional reasoning about stress, Phase II used focus groups built around a playful probe. Playful probes extend the design-probe tradition by embedding the inquiry in a game-like artifact that participants enact, producing data that surfaces through play rather than through direct questioning (**bernhaupt·playful·2007; jarke·probes·2018**). This format fits the research aim because the mechanics that mattered most in the Phase I data were those associated with narrative authorship, and a probe that asked participants to author

stress narratives themselves would let us observe how those mechanics operate in use.

We adapted *Fiasco* (**morningstar·fiasco·2009**), a GM-less tabletop role-playing game in which players improvise a short story from a customizable setting (a *play-set*) and a set of relationships, needs, locations, and objects. We chose *Fiasco* for three reasons: its playsets are fully customizable, which allowed us to replace the default setting with one grounded in student experience; its GM-less structure distributes narrative authority across all players, which matches the codesign aims of the study; and its improvisational format asks players to act from within a character’s situation rather than describe it from outside, which surfaces coping reasoning as an embedded response rather than as an abstract report. Role-playing games have been argued to support reflection on stress and mental health through this same mechanism of embodied perspective-taking (**ian’s·baker·role-play·2022**; **yuliawati·scoping·2024**).

The custom playset, “Blackthorn University,” placed players in a fictional institution where a murder mystery unfolded alongside ordinary academic pressure. The playset was developed by adapting the community *Fiasco* playsets *UC Hellmouth* and *Town and Gown*, both of which situate *Fiasco*’s interpersonal-drama mechanics within a university setting, and extending them with a custom card deck of Characters, Relationships, Needs, Locations, Objects, Kickers, and Tilts. Each card was written to foreground a stress theme grounded in research on college-student mental health. For example, *The Scholar* character card encodes academic burnout; *The Cursed Mentor and the Doomed Student* relationship card encodes mentorship pressure and imposter syndrome; the *Midnight Bridge* location card encodes fear of the unknown; and Kicker cards such as *The Exam That Was Never Scheduled* and *The Time Loop Incident* encode performance anxiety and loss of control, respectively. Players did not receive prescribed coping mechanisms. Instead, they improvised responses to these framings in character, which allowed us to observe how coping strategies emerged, combined, and broke down during stressed play.

Eleven focus groups were conducted on the University of Minnesota Duluth campus with a total of thirty undergraduate and graduate students ($N = 30$, ages 18–30, $M = 22.4$). Participants were recruited via campus flyers, and each received \$20 compensation for their time. Because the Phase I survey was anonymous, we cannot determine whether any Phase I respondents also participated in Phase II; participation in each phase was

treated independently, and no identifiers linked the two data sets. Group sizes ranged from two to four participants, reflecting the Fiasco ruleset’s intended player count rather than conventional focus-group norms. This phase was approved by the Institutional Review Board. Sessions were facilitated by one of three members of the research team, each of whom followed a shared facilitator script to maintain procedural consistency across groups.

Each 90-minute session followed a three-act structure:

Table 3.1: Phase II Narrative Focus Group Session Structure

Act	Duration	Activity	Data Captured
Act 1: Narrative Roleplay	30 min	Collaborative storytelling set at “Blackthorn University,” guided by a facilitator. Participants played characters equipped with common coping mechanisms and drew from custom prompts reflecting stressors identified in prior research, such as imposter syndrome and time pressure.	Audio-recorded role-play dialogue; facilitator notes on collective story construction
Act 2: Post-Game Interview	20 min	Semi-structured reflection on the alignment between each character’s in-game stress responses and the participant’s real-world coping strategies. Facilitator probed for which mechanics felt evocative and which felt inert.	Interview transcripts on coping resonance and mechanic-level response

Act	Duration	Activity	Data Captured
Act 3: Co-Design Activity	30 min	Critique of existing wellness applications (<i>Finch</i> , <i>Headspace</i>) followed by participant-led sketching of wireframes for a narrative-based alternative.	Wireframe sketches; verbal design rationale captured in session audio

Note. The three acts above account for 80 minutes of structured activity within the 90-minute session. The remaining 10 minutes were used for introductions, informed-consent review, and closing debrief across the start and end of each session.

Sessions were audio-recorded with participant consent. Transcripts, facilitator notes, and participant sketches were analyzed using reflexive thematic analysis, as detailed in Section 3.7.

3.4 System Design and Development

In line with RtD methods, this stage translated the findings of Phases I and II into a working artifact, *The Archive*, which would then be tested through expert evaluation and field deployment. This stage is the central site of RQ3, which asks how a codesign approach informs the development of narrative-centered systems for stress reflection. The Phase I and Phase II findings did not specify a design; they identified constraints and opportunities that the design had to respond to. This section reports the design decisions that followed from those constraints and the shape of the artifact that resulted. A fuller account of the system and its rationale is given in Chapter ??.

Design principles drawn from Phases I and II

Four principles emerged from the synthesis of Phase I and Phase II data.

First, *narrative over task*. Phase I rated story-driven quests and personalization as the most preferred mechanics, and rated leaderboards, streak calendars, and explicit progress tracking as the least preferred. Phase II confirmed this in use: when participants critiqued existing wellness applications, they described streak-based designs as

chore-like and motivating only until completion. The Archive was therefore built around narrative progression rather than task completion.

Second, *choice without penalty*. Phase II participants consistently rejected game systems that punished failure or produced guilt. Narrative choices in The Archive have consequences within the story but are not graded, scored, or penalized. Reflection activities are optional and embedded in the narrative rather than required.

Third, *guidance without instruction*. Phase II revealed a split between participants who flourished in open-ended creative play and those who wanted a more directive voice to structure the experience. To accommodate both, The Archive includes two NPCs: The Curator, a non-directive guide who frames each module and responds to player choices without prescribing outcomes, and Poppy, a peer-anchored character introduced in the third module who models coping strategies from within the narrative rather than from outside it.

Fourth, *spatial and symbolic framing*. Phase II participants consistently used spatial metaphors to describe stress: haunted libraries, corridors that shifted, spaces that vanished. The Archive renders its six modules as environments the player traverses rather than tasks the player completes, and it represents progress through symbolic items rather than through visible scores or percentages.

The resulting artifact

The Archive is a narrative-centered mobile application in which the player assumes the role of a reader-protagonist working through a sequence of bound volumes. The application contains six modules, each addressing a distinct dimension of academic stress: *Prologue* (orientation and tutorial), *Embark* (imposter syndrome, framed through a decaying library), *Lounge* (obligation and unreadable contracts), *Poppy* (emotional vulnerability and peer support), *Focus* (anxiety under pressure, framed through a crumbling environment to reconstitute), and *Contract Fulfilled* (Jungian shadow work and integration of repressed stress).

Each module follows a choice-based narrative loop. Passages load at a pace the player selects (slow, medium, or fast), and reach choice points where the narrative pauses until the player selects a response. Two embedded activities are distributed across the

modules. *Inscriptions*, a narrative journaling feature, appears as a prompt at moments determined by player choices and accepts free-text reflection without evaluation. *Mind Defender*, a minigame with dynamic difficulty adjustment, appears in modules addressing cognitive overload and frames thought-reframing as an active gameplay task. Progress is represented through *Lumen*, a symbolic reward system that accumulates through engagement rather than through measured achievement.

The design deliberately departs from several conventions that participants' wireframes borrowed from commercial gamified DMH applications, including cozy decorative rooms, pet companions, gear loadouts, and level-up awards. These conventions were legible to participants but were associated with the chore-and-reward patterns that Phase II had identified as least supportive. The first iteration of The Archive therefore prioritized narrative depth and interpretive engagement over these visual affordances, with the expectation that subsequent iterations could reintroduce aesthetic elements once the narrative foundation was stable.

Technical implementation

The application was developed with a React Native frontend targeting Android for the field deployment, paired with a Django backend and a PostgreSQL database that stores narrative state, journal entries, and player data. Narrative content was authored in Arcweave, which supports the complex branching and state-dependent passages that *The Archive*'s structure requires. Figure 3.2 shows the Arcweave authoring board for the *Embark* module as an illustrative sample of the narrative structure. The system architecture, module-by-module narrative content, and full interaction flow are detailed in Chapter ??.

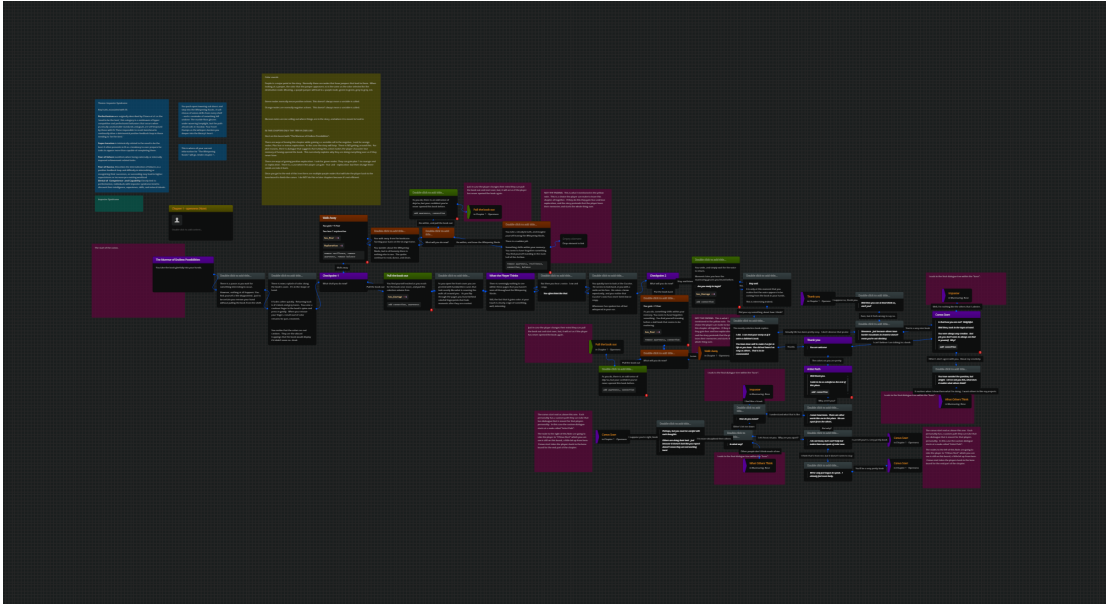


Figure 3.2: Arcweave authoring board for the *Embark* module (Chapter 1, *Openness*), which addresses imposter syndrome through the metaphor of a decaying library. Individual passage text is not intended to be legible at this scale. The figure is included to show the structural complexity of the authored narrative, including passage cards, branching player choices, canvas-to-canvas jumpers, and the three terminal paths (*Imposter*, *Artist Path*, and *What Others Think*) through which the module resolves.

3.5 Expert Evaluation and Design Validation

Following the first iteration of the artifact, we conducted a multi-disciplinary expert evaluation to identify usability issues, assess therapeutic and narrative alignment, and inform design refinements before field deployment. This stage serves RQ3 by providing a second round of expert-grounded input into the iterative design process, between the codesign work of Phase II and the naturalistic evaluation of Phase III. We adopted a cross-disciplinary rather than single-field review, on the basis that interactive systems at the intersection of HCI, narrative design, and mental health benefit from evaluators who reflect the full range of disciplines the artifact engages. Paz and colleagues show that heuristic evaluation by specialists from related fields can produce results comparable in

reliability to evaluation by HCI specialists alone, which warrants the multi-disciplinary approach adopted here (**paz·applying·2023**).

Ten faculty and staff at the University of Minnesota Duluth were recruited by direct email invitation from the research team, based on disciplinary expertise relevant to different facets of *The Archive*’s design. Their expertise spanned five areas: gamification and games (E1, E3, E9), creative writing and rhetoric (E2, E4, E8, E10), mental health counseling (E7), world languages and cultures (E5), and social sciences (E6). Participation was unpaid, consistent with standard practice for collegial expert review within a single institution. Experts are identified throughout this thesis by assigned codes (E1–E10) to protect anonymity. Table 3.3 summarizes the expert participants by department and area of expertise. This phase was approved by the Institutional Review Board.

Table 3.3: Expert participant overview ($N = 10$). Participants are identified by assigned code. Department and expertise reflect the areas relevant to the prototype’s interpretation during evaluation.

ID	Department / Affiliation	Expertise
E1	Electrical Engineering	Gamification
E2	Writing	Creative Writing
E3	Computer Science	VR Games
E4	Library	Creative Writing
E5	Interdisciplinary Studies	World Languages & Cultures
E6	Philosophy	Social Sciences
E7	University Health Services	Counselor
E8	Writing	Rhetoric
E9	Computer Science	Games in Education
E10	Writing	Advanced Writing

Each evaluator completed a single one-hour session held in the evaluator’s office or workplace. At the start of each session, participants were introduced to the study goals and asked for verbal consent to recording. Each participant was then issued fresh login credentials so they could walk through the application as a first-time user using the

Character Creator onboarding pathway, beginning from account creation.

The session combined a think-aloud walkthrough of the application with a usability survey and a semi-structured interview. During the walkthrough, evaluators were asked to work through the onboarding pathway, the tutorial module, a narrative module with a branching choice point, the Inscriptions journaling activity, and the Mind Defender minigame. They were encouraged to verbalize their reactions to each task as they worked. After completing the task set, evaluators were given unstructured time to explore the remainder of the application. Evaluators then completed a ten-item usability survey administered in Qualtrics, and the session concluded with a semi-structured interview. The interview was organized into a general block asked of all evaluators and four domain-specific blocks tailored to the evaluator’s expertise: usability and HCI, game design, mental health, and creative writing. The full evaluation protocol, including the survey items and interview prompts, is reproduced in Appendix ??.

All ten sessions were facilitated by the lead researcher to maintain consistency of prompting across evaluators, and audio-recorded with participant consent. Findings from this phase shaped the second iteration of the artifact, in particular the prioritization of the story modules in the interface and the refinement of narrative pacing, as described in Section 3.6.

3.6 Phase III: Student Deployment and Use Evaluation

The final phase examined *The Archive* in a naturalistic context over a five-day deployment followed by an in-person post-evaluation. This phase serves all three research questions: it produces pre/post change data relevant to RQ1, game-experience ratings and exit-interview accounts relevant to RQ2, and a full-cycle demonstration of the code-designed artifact in use relevant to RQ3. This phase was approved by the Institutional Review Board.

Before deployment, we refined the artifact using findings from the expert evaluation. Evaluators had flagged a disconnect between the structure of the UI and the story, so we restructured the interface to foreground the story modules rather than the ancillary features. This decision aligned the system more closely with narrative-forward applications such as Betwixt (**kitromili’interactive’2025**) and away from the chore-and-reward

structure of applications such as Finch. Because evaluators saw the strongest promise in the act of working through stress events through a narrative persona rather than directly, we prioritized the module experience in the final build. Several experts also emphasized the need for stronger aesthetics to support students’ mental model of the story world. These iterative changes allowed us to omit features that risked distracting from the narrative and to strengthen the story content itself.

The application continued to be developed in React Native and was distributed as an Android APK via Expo Application Services. Students who signed up for the study received a QR code linking to a scheduler, through which they booked a 30-minute in-person onboarding session. Onboarding consisted of a brief pre-survey, an overview of the application and the research, installation of the APK on the participant’s device or on a loaner device provided for the study, and a guided walkthrough of the core features. During this session, participants were issued login credentials, created their avatar, selected a time for daily reminder notifications, and played through the *Prologue*, which serves as the application’s tutorial module. Participants who did not own an Android device were issued a loaner for the study duration. Participants were recruited via campus flyers at the University of Minnesota Duluth and received \$40 compensation for their time.

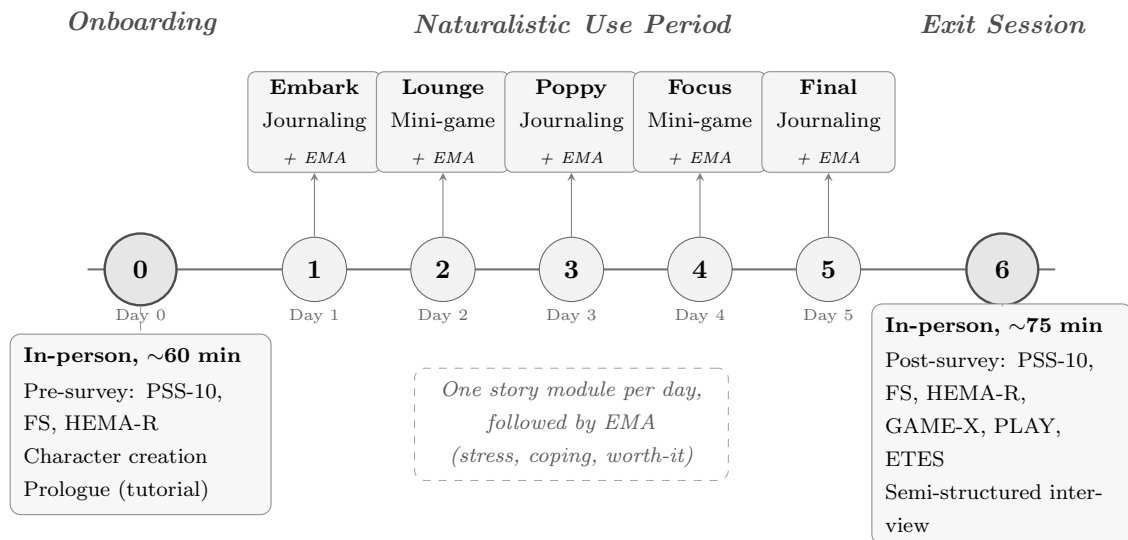


Figure 3.3: Phase III deployment timeline. Day 0 and Day 6 were in-person research sessions bracketing a five-day naturalistic use period. On each of Days 1–5, participants engaged with one story module at a self-selected time, followed by a three-item Ecological Momentary Assessment (EMA) capturing stress, coping confidence, and perceived worth. Pre-deployment instruments were administered during onboarding; post-deployment instruments and the semi-structured interview were administered during the exit session.

The five-day deployment followed the onboarding session. Participants were asked to complete one story module per day, progressing through *Embark*, *Lounge*, *Poppy*, *Focus*, and *Contract Fulfilled* in sequence. Each module contained an embedded reflective activity, either an Inscriptions journaling prompt or a Mind Defender minigame episode. To prioritize autonomy, the system did not enforce a minimum session length and allowed participants to engage at times of their own choosing. A daily reminder notification at the participant’s selected time prompted them to attempt the day’s assigned module.

On Day 6, participants returned for a 75-minute in-person post-use session. This session repeated the pre-survey measures, added three post-only instruments specific to the user’s experience of the application, and concluded with a semi-structured exit interview organized around the three research questions. Data was collected across the full phase through four primary channels:

1. **Pre- and Post-Surveys:** The pre-survey, administered during onboarding, included the Perceived Stress Scale (PSS-10) (**cohen’global’1983**), the Flourishing Scale (FS) (**diener’new’2010**), and the Hedonic and Eudaimonic Motives for Activities scale, revised (HEMA-r) (**huta’pursuing’2010**). The post-survey, administered on Day 6, repeated these measures and added three instruments specific to the user’s experience of the application: the Game Experience Questionnaire (GAME-X) (**law’systematic’2018**), from which we report the Absorption and Agency subscales; the PLAY Heuristics for player experience (**desurvire’game’2009**); and the Eudaimonic Technology Experience Scale (ETES) (**wozniak’quantifying’2023**), from which we report the Eudaimonic Goals and Self-Knowledge factors. Pre- and post-survey responses were matched via a self-generated identifier composed of the

first letter of the participant’s mother’s first name, the first letter of the participant’s last name, and the last four digits of the participant’s phone number. All eighteen enrolled participants produced a matching identifier across both surveys and are included in the matched pre/post analyses reported in Chapter 5.

2. **Ecological Momentary Assessment (EMA):** Upon completing each of the five story modules, participants were shown a three-item survey assessing in-the-moment stress (*stress_now*, 0–10), coping confidence (*coping_confidence*, 1–10), and perceived worth of the session (*worth_it*, 1–5).
3. **Interaction Logs:** The application logged narrative choices, journal entries, and minigame engagement to contextualize the self-reported data.
4. **Day 6 Semi-Structured Exit Interview:** Following the post-survey, each participant completed an in-person semi-structured interview of approximately 45 minutes. Questions were organized into three blocks, each aligned with one research question. The RQ1 block asked about changes in stress awareness, stress response, and memorable story moments during the deployment. The RQ2 block asked about the branching-choice structure, the journaling and minigame activities, the Lumen progression system, and the experience of story resolution. The RQ3 block asked about the perceived authenticity of the stress scenarios and the narrative-first design compared with other wellness applications. The full interview protocol is reproduced in Appendix C.4.

Analysis of these four data sources is described in Section 3.7, and the full findings are reported in Chapter 5.

3.7 Data Analysis

The analytic strategy matched the role of each data source within the overall design. Phase I produced rating data used to characterize student preferences and to inform later design decisions, so it was analyzed descriptively. Phase III produced matched pre/post measures, post-survey game-experience measures, daily EMA responses, interaction logs, and exit interview transcripts. These were analyzed to address RQ1 (change

in perceived stress and wellbeing over the deployment), RQ2 (the relationship between specific qualities of the game experience and outcomes), and RQ3 (how the codesigned artifact performed in use and what participants said about it).

Quantitative data from Phase I was analyzed using descriptive statistics for all Likert-scale items, with means and standard deviations calculated per mechanic. Given the sample size ($N = 38$) and the exploratory role of this phase, descriptive summaries were reported without inferential testing. For the open-ended responses, word clouds were generated to surface recurring terms, and free-text responses were read alongside the quantitative ratings to contextualize them. For category-level analysis, items that were conceptually related were grouped, and a category mean was calculated by averaging the constituent item ratings. The groupings were organized through the Mechanics–Dynamics–Aesthetics (MDA) framework (**hunicke'mda'nodate**). MDA is used here as the common analytic vocabulary across both the board-game probe of Phase II and the mobile application of Phase III, and it is also the framework Deterding and colleagues draw on when defining gamification in terms of game design elements and their resulting player experience (**sebastian'deterding'game'2011**). Using MDA consistently across both mediums allowed the survey evidence, the focus-group observations, and the later system-design decisions to be compared on the same terms.

Quantitative data from Phase III was analyzed using descriptive statistics and change-score analysis ($\Delta = \text{Post} - \text{Pre}$) for the three measures administered at both time points (PSS-10, FS, HEMA-r). Per-measure means, standard deviations, and change-score distributions were computed across all eighteen matched participants. Pearson correlations were computed between the post-survey game-experience measures (GAME-X Absorption and Agency, PLAY Heuristics, ETES Eudaimonic Goals and Self-Knowledge) and the wellbeing outcomes, to examine whether specific qualities of the game experience were associated with pre/post changes. Given the sample size, these correlations are reported as descriptive patterns rather than as inferentially tested associations; interpretation of these relationships is accordingly cautious and framed against the small sample.

The EMA data were analyzed as daily trajectories across the five-day deployment, with per-day means computed for each of the three items (*stress_now*, *coping_confidence*,

worth_it) and inspected for patterns of change across the module sequence. EMA compliance was partial rather than complete. The EMA was designed to appear immediately after each story completion, but two factors reduced the number of submitted responses. Some participants closed the prompt inadvertently before submitting, and a bug in the trigger logic suppressed the prompt on a subset of completions. For this reason, EMA responses were analyzed as they were received (n=47 responses across the eighteen participants) rather than as a balanced panel. Reliability at the response level was computed across the three items to confirm that they behaved coherently as a within-session assessment, and is reported in Chapter 5.

Interaction logs were used as contextual reference material rather than as a standalone analytic dataset. Aggregate patterns such as completion counts, choice distributions, and minigame engagement were inspected to corroborate self-reported engagement patterns and to identify anomalies (for example, participants who reported finishing a module but whose logs showed incomplete progression). Where such anomalies affected analysis decisions, they are noted in Chapter 5.

Qualitative data from Phase II, the expert evaluation, and the Phase III exit interviews were analyzed using reflexive thematic analysis (**braun'using'2006; braun'reflecting'2019**). Audio recordings were transcribed using Glean, checked against the originals, and coded inductively through a process of familiarization, initial code generation, theme construction, and review. For Phase II, themes were interpreted through the MDA framework to examine the relationship between design structure and student stress interpretation. For the expert evaluation, themes were organized around the system's usability, narrative coherence, and aesthetic alignment with the stress-management framing. For the Phase III exit interviews, analysis followed a participant-level memoing approach (**byrne'worked'2022**): an analytic memo was produced for each of the eighteen participants who completed an exit interview, consolidating their account across the three research-question blocks before themes were constructed across cases. This approach preserved the integrity of individual accounts during early analysis and produced an audit trail that connected each theme to specific participant extracts. The findings from all analyses are reported in Chapter 5.

3.8 Ethical Considerations

All study phases were approved by the University of Minnesota Duluth Institutional Review Board prior to deployment. Participation was voluntary across all phases, and participants were informed that they could withdraw at any time without consequence and without providing a reason. Consent was obtained through a Qualtrics-based consent form in Phases I and III, and through verbal consent during the expert evaluation.

Because this study engaged stress-related content, we took several precautions against participant harm. In Phase II, participants were given “let’s not” cards during the Fiasco session and could play one to indicate that they did not wish to engage with a particular topic, after which the narrative was redirected. In the deployed application, each story module includes a content warning button that the participant can tap before entering the module to see the themes covered in that chapter, and every passage allows the participant to pause, leave, or return to a neutral hub. Participants were told during onboarding that they could skip sensitive narrative content, decline to complete a reflective activity, or end a session at any point without affecting their compensation. The application also lists campus and national mental health resources, including the University of Minnesota Duluth Health Services counseling line and the 988 Suicide and Crisis Lifeline, and these resources are reachable from within the application at any time.

Participant data was collected under a pseudonymization strategy rather than full anonymization. Phase I survey responses were collected anonymously. Phase II focus groups were audio-recorded with participant consent and the recordings were transcribed and de-identified for analysis. Phase III required identifiable data during the study itself, because participants used personal accounts within the application and because pre- and post-surveys needed to be matched to the same person. For this matching we used a self-generated identifier composed of the first letter of the participant’s mother’s first name, the first letter of their own last name, and the last four digits of their phone number. This approach avoided the collection of names, email addresses, or other direct identifiers during analysis. Within the thesis and any associated outputs, participants are identified only by assigned codes (PXX–PXX for Phase III participants and EXX–EXX for expert evaluators). Audio recordings, transcripts, and interaction logs are

stored in secure, access-controlled university storage and will be retained for five years following study completion, after which identifiable data will be destroyed.

Compensation was structured to acknowledge participants' time without creating inappropriate inducement. Phase I was unpaid, consistent with standard practice for short screening-style surveys. Phase II participants received \$20 for a 90-minute session. Phase III participants received \$40 for the full study, covering the onboarding session, five days of use, and the Day 6 post-use session. Expert evaluators were unpaid, consistent with standard practice for collegial review within a single institution. Participants were informed that their compensation would not be reduced if they skipped sensitive content, declined a task, or withdrew from the study.

We also considered ethical aspects of the research context. The study recruited exclusively from a single institution at which the lead researcher was a graduate student, and we did not recruit from any class in which the researcher held a teaching or grading role, to avoid any coercive overlap between the study and academic evaluation. The research team reflected openly throughout the project on the tension between asking participants to engage with stress-related content and the study's goal of evaluating a wellness-oriented artifact. This tension motivated the design choices described in Section 3.4, in particular the commitment to optional reflection, non-penalty choice, and the presence of an NPC guide who frames rather than evaluates the player's decisions.

Chapter 4

System Design

4.1 Design Intent and Guiding Principles

The Archive was designed in response to findings across Phases I and II. The initial survey indicated that students rated narrative and personalization-oriented mechanics most favorably. Conversely, competitive mechanics such as streaks, leaderboards, and performance tracking were described as pressure-inducing or discouraging. Focus group sessions reinforced these patterns; students consistently rejected guilt-based feedback and described existing wellness applications as feeling like chores or being disconnected from the reality of stress. Participants utilized spatial and environmental metaphors to describe their experiences, including haunted libraries, looping corridors, and vanishing spaces. When imagining alternatives, they requested emotionally safe structures, optional reflection, and choice without penalty.

These findings shaped a core set of design values reasoned through the Mechanics-Dynamics-Aesthetics (MDA) framework (**seaborn'gamification'2015**). The MDA framework distinguishes between structural rules (mechanics), the behaviors those rules produce during interaction (dynamics), and the experiential qualities the player feels (aesthetics). We utilized this framework to work backward from the aesthetic qualities students identified as meaningful: emotional safety, agency, and reflection without pressure.

Four primary mechanic-level decisions followed from this process. First, narrative choice replaces task completion as the primary structural mechanic to foster a dynamic

of authorship and an aesthetic of agency. Second, progress is represented through narrative movement and symbolic items rather than scores or visible metrics. This responds directly to the focus group critique of static progress bars as motivating only until completion. Third, reflection is optional and embedded within the story flow. This ensures the application does not become another obligation for the student. Fourth, guidance is delivered through character dialogue rather than system instruction, responding to the user need for a companion that contextualizes rather than evaluates.

4.2 Technological Overview

The application was built for mobile deployment using React Native on the frontend to enable cross-platform support. The backend was constructed in Django to provide application APIs, with PostgreSQL utilized to store story content, user data, journal entries, and game state.

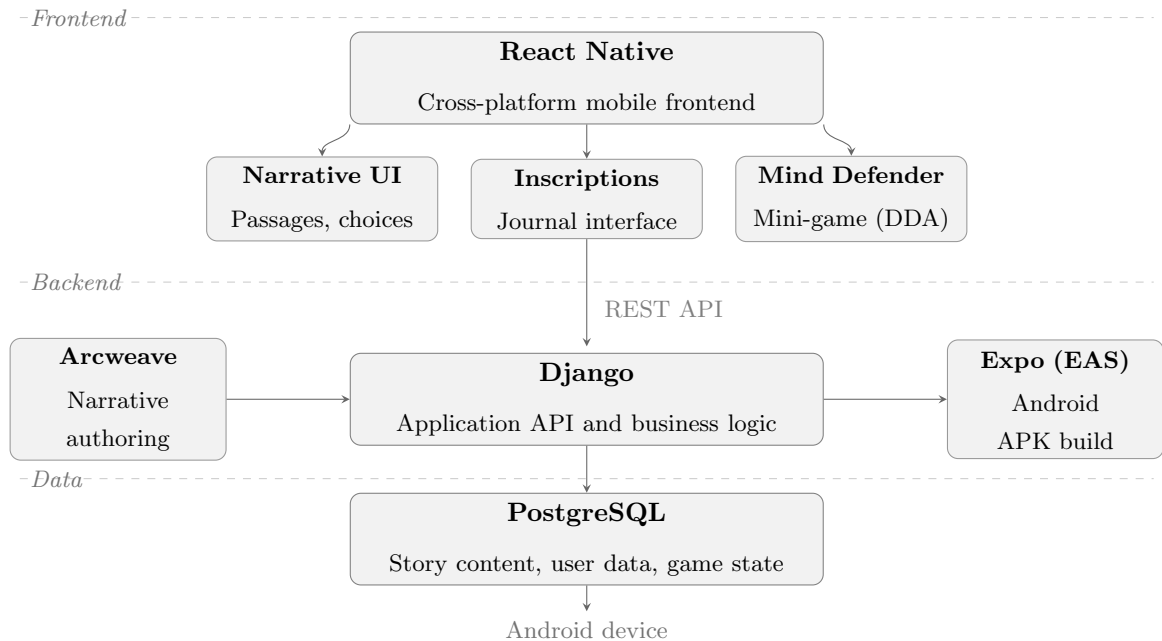


Figure 4.1: System architecture of The Archive, showing the React Native frontend, Django backend API, PostgreSQL data layer, Arcweave narrative authoring pipeline, and Expo deployment.

Narrative content was authored using Arcweave, which allowed the complex branching logic developed through participatory work to be mapped directly into the application interaction flow. For deployment during the expert evaluation and Phase III, the application was packaged using Expo Application Services and distributed as an Android APK.

4.3 Core Interaction Loop and Narrative Structure

The interaction loop centers on a choice-based narrative system. Passages load automatically at a pace selected by the player, who chooses between slow, medium, and fast reading speeds. This pacing system responds to focus group feedback emphasizing autonomy over the rhythm of engagement.

Passages are presented according to their narrative function. Background narration appears as white text on the screen to establish scene context. Dialogue from non-player characters (NPCs) appears in text bubbles alongside character images. When a choice point is reached, the passage pauses until the player selects a response. After a choice is made, the player's avatar appears beside a reply bubble to ground the decision within their narrative identity.

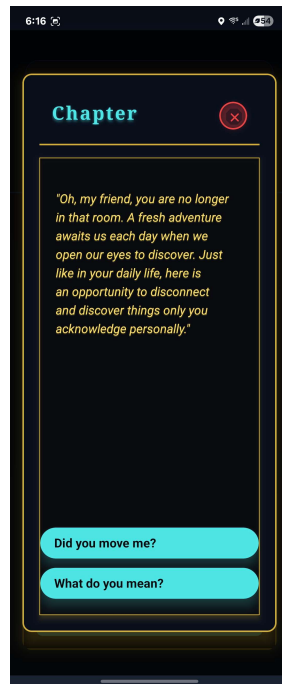


Figure 4.2: Passage rendering in Version 1 (pre-evaluation), showing background narration in white text.

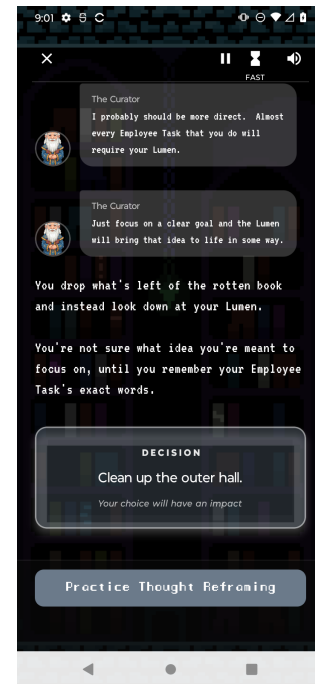


Figure 4.3: Passage rendering in Version 2 (post-evaluation), showing refined text contrast and layout.

Each of the five story modules addresses a distinct thematic dimension of academic stress:

- **Embark:** Addresses imposter syndrome through the metaphor of a decaying library.
- **Lounge:** Explores confusion regarding obligations and the weight of unreadable contracts.
- **Poppy:** Focuses on emotional vulnerability and peer-anchored coping advice.
- **Focus:** Manages anxiety under pressure, requiring the player to reconstitute a crumbling environment.
- **Contract Fulfilled:** Utilizes Jungian shadow work to encourage the integration of repressed stress.

4.4 Inscriptions: Narrative-Embedded Journaling

Inscriptions is the primary reflective feature of the application. Rather than existing as a separate section, journal prompts are embedded directly into story modules at moments determined by player choices. This keeps reflection tied to the narrative experience rather than framing it as a separate assessment task.

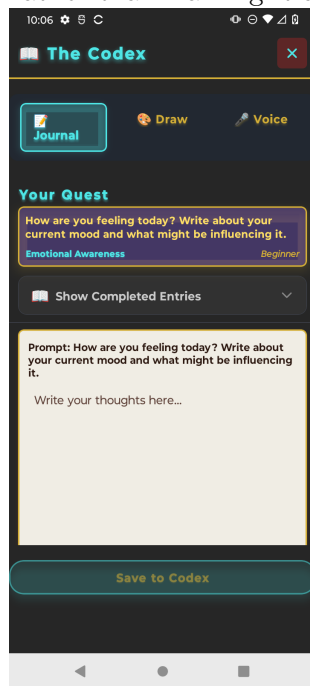


Figure 4.4: Inscriptions journal interface in Version 1 (pre-evaluation).

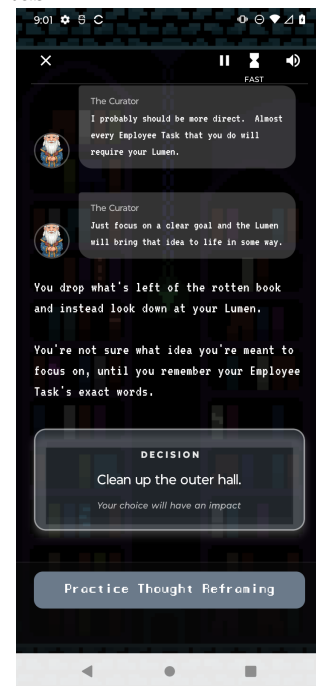


Figure 4.5: Inscriptions journal interface in Version 2 (post-evaluation), reflecting expert feedback on prompt clarity and visual contrast.

Each prompt is presented as a written text entry, a drawing, or a voice memo. The format is determined by the specific path the player took through the story. For example, in the *Embark* module, a player who notices the stillness of the library receives a drawing prompt, while a player who notices the smell receives a voice prompt. This pairing ensures the mode of expression feels connected to the narrative moment. Prompts use the story’s metaphors to invite reflection, such as asking what “unfinished” work looks like in the player’s own life.

4.5 Mind Defender: Active Thought Reframing

Mind Defender is an optional mini-game that offers an active mode of engagement alongside narrative and journaling. The game is a 2D action experience where players use a bow to defend against incoming enemies that symbolically represent negative thought patterns. When an enemy is destroyed, it reveals a thought statement drawn from the Automatic Thoughts Questionnaire. collectible power-ups reveal positive thought counterparts that provide temporary gameplay advantages.



Figure 4.6: Mind Defender mini-game in Version 1 (pre-evaluation), showing the 2D action gameplay and incoming thought-pattern enemies.

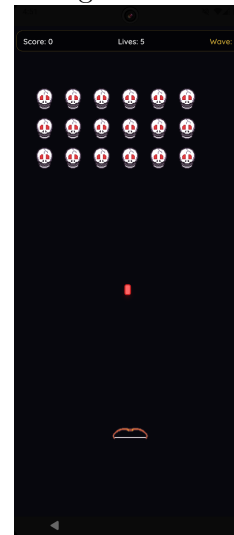


Figure 4.7: Mind Defender mini-game in Version 2 (post-evaluation), reflecting expert feedback on visual clarity and difficulty pacing.

The mini-game utilizes a Dynamic Difficulty Adjustment (DDA) system to manage emotional intensity. The system monitors behavioral indicators including hit frequency, accuracy, and survival intervals. When a pattern suggesting interaction strain is detected, the system reduces enemy speed and increases the probability of supportive power-ups. This creates a stabilization window that allows the player to recover without the experience ending abruptly. This prioritization of support over challenge reflects the core design value of maintaining engagement during periods of high stress.

4.6 Supporting Characters: The Curator and Poppy

The Archive includes two NPCs designed to offer guidance without authority. The Curator is the primary guide and functions as a narrative scaffold. He explains the logic of the world but does not evaluate the player’s behavior or prescribe correct actions. His non-directive dialogue aligns with focus group feedback expressing discomfort with parental or corrective digital companions.

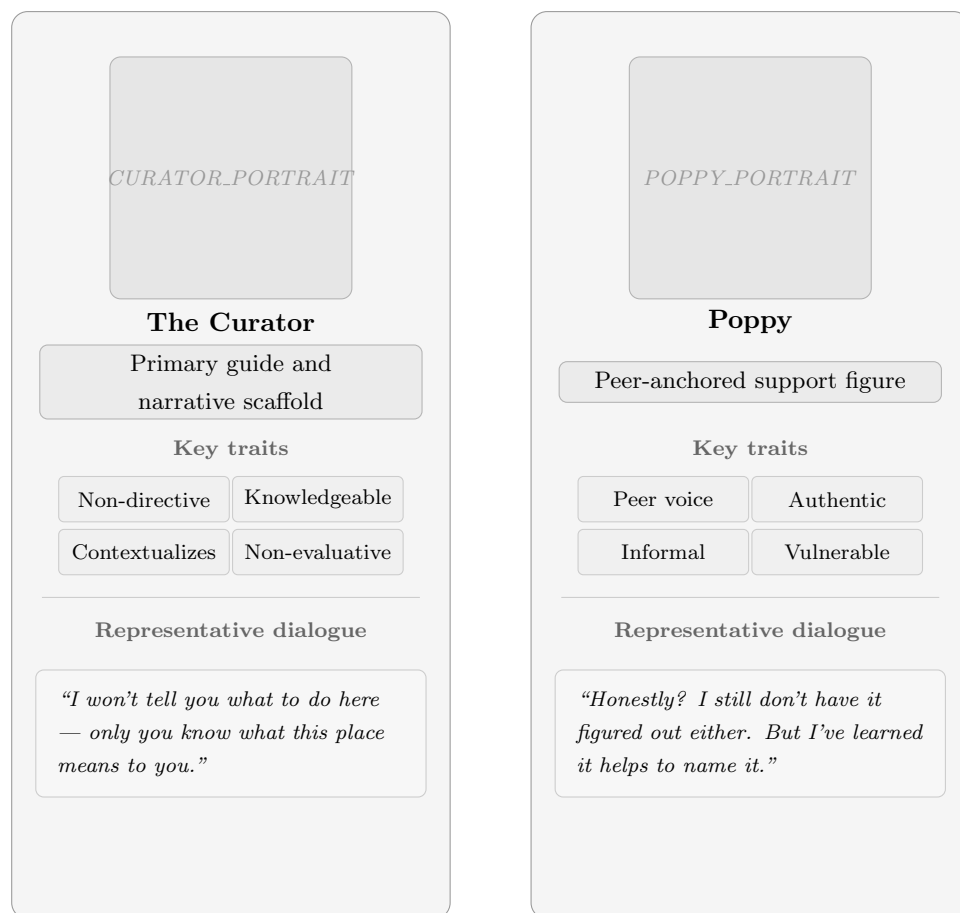


Figure 4.8: Character personas for The Archive’s two NPCs. The Curator serves as a non-directive narrative guide; Poppy operates as a peer-anchored support figure.

Poppy is introduced in the third module as a peer-anchored support figure. She speaks from personal experience rather than clinical expertise, admitting to her own

struggles with the creature. Her office is cluttered and she is dressed in pajamas, signaling an authenticity that participants identified as more meaningful than performative wellness instruction. Together, these characters operationalize the principle that supportive guidance should feel like a knowledgeable companion rather than an evaluative system.

Chapter 5

Findings

5.1 Overview

This chapter presents findings across four phases of the study, organized as a design research sequence moving from preference discovery through participatory design, prototype evaluation, and deployment. Phase I (§5.2) reports results from a cross-sectional survey ($N = 38$) establishing how college students rate gamification mechanics and stress-management features in digital mental health applications. Phase II (§5.3) presents thematic findings from eleven participatory focus groups ($N = 30$) in which students used narrative roleplay to articulate their experiences of stress and their expectations for wellness design. The Expert Evaluation (§5.4) reports results from six think-aloud sessions with domain experts who assessed *The Archive* prior to deployment. Sections 5.5–5.12 present quantitative findings from Phase III, the five-day deployment of *The Archive* with $N = 18$ enrolled participants.

Each phase built directly on the one before it. Phase I established what students wanted from gamified wellness tools and what they rejected. Phase II surfaced how students understand and narrate stress, translating those accounts into concrete design requirements through co-design activities. The Expert Evaluation provided domain-specific critique of the prototype before participants encountered it, identifying usability and therapeutic alignment issues that were addressed prior to deployment. Phase III then assessed whether the resulting system, shaped throughout by this iterative process, produced measurable wellbeing effects and positive experiential qualities under

naturalistic engagement conditions.

Phase III collected data from three sources. Pre- and post-surveys assessed wellbeing outcomes using the Perceived Stress Scale (PSS-10), Flourishing Scale (FS), and the Hedonic and Eudaimonic Motivation for Activities scale (HEMA-r). Daily Ecological Momentary Assessment (EMA) surveys captured in-the-moment stress, coping confidence, and perceived worth of engagement immediately following each story module. Post-only game experience scales — GAME-X, PLAY Heuristics, and the Eudaimonic and Transformative Experience Scale (ETES) — assessed participants’ perceptions of the system’s design qualities. Scale descriptions and scoring procedures are provided in the Methods chapter; relevant ranges and interpretive thresholds are noted where needed in the text below.

Fifteen of the 18 enrolled participants were successfully matched across pre- and post-surveys using the unique identifiers collected at enrollment. Three participants who enrolled on April 2, 2026 did not complete the post-survey within the study window and are excluded from matched analyses; their pre-survey profiles are noted where relevant. One test response was removed during data cleaning. A total of 47 EMA responses were collected from 18 unique users across the five-day period. Because this study was exploratory and lacked a control group, all findings are interpreted as preliminary trends rather than causal claims. Limitations are discussed in Section 5.12.

5.2 Phase I Findings: Experiential Survey

5.2.1 Sample and Engagement with Existing Tools

Thirty-eight participants completed the Phase I survey, with item-level valid responses ranging from $N = 24$ to $N = 37$. The sample comprised 21 male-identifying, 16 female-identifying, and 2 non-binary-identifying participants across academic levels (21 graduate students, 7 seniors, 6 freshmen, 3 juniors, 2 sophomores). Of the 40 respondents, 14 currently used at least one self-care application; 25 did not. Open-text responses from non-users described existing applications as impersonal, commercially motivated, or too generic. One participant reflected:

“I guess there’s something inhuman about an app that puts me off. Without

an in-person connection with a therapist or something, I find it harder to shake the feeling that someone is trying to sell me something. I like the idea of designing games to be relaxing and stuff; like, Journey was an extremely memorable experience for me.”

This skepticism toward the existing tool landscape grounded the rationale for an alternative approach and is consistent with the coping activity data reported in §5.2.5.

5.2.2 Gamification Mechanic Preferences

Participants rated five core gamification mechanics on a 1 (*one of the worst*) to 5 (*one of the best*) scale. As shown in Table 5.1, story-driven quests received the highest mean rating ($M = 3.56$, $SD = 1.46$), followed by personalization ($M = 3.47$, $SD = 1.54$) and collaborative activities ($M = 3.16$, $SD = 1.61$). Progress bars fell below the midpoint ($M = 2.97$, $SD = 1.36$), and points and leaderboard mechanics received the lowest rating of the five ($M = 2.16$, $SD = 1.48$) — more than a full scale point below narrative features.

Table 5.1: Student ratings of core gamification mechanics (1 = *one of the worst*, 5 = *one of the best*).

Game Design Mechanic	M	SD	N
Story-driven quests	3.56	1.46	36
Personalization features	3.47	1.54	36
Collaborative activities	3.16	1.61	37
Progress bars / leveling systems	2.97	1.36	36
Points / leaderboard mechanics	2.16	1.48	37

The gap between the distribution’s poles is more than a scaling artifact: narrative and personalization mechanics outscored leaderboards by more than a full point, suggesting a categorical divide rather than a matter of relative preference. Students’ resistance was most sharply directed at competitive and comparative features; collaborative and progression-based mechanics attracted moderate endorsement, indicating that game-based structure per se was not the problem.

5.2.3 Mood and Stress Relief: Category-Level Patterns

A second set of nine items asked whether specific game elements help improve mood or reduce stress (1 = *strongly disagree*, 5 = *strongly agree*). Results are shown in Table 5.2. Engaging stories led ($M = 3.58$), followed by exploration ($M = 3.44$), relaxing visuals and humor (both $M = 3.35$), and collection mechanics ($M = 3.22$). Physical activity received the lowest endorsement ($M = 2.14$).

Table 5.2: Student ratings of game elements for mood improvement and stress reduction (1 = *strongly disagree*, 5 = *strongly agree*).

Game Element	<i>M</i>	<i>SD</i>	<i>N</i>
Engaging stories	3.58	1.52	36
Exploration	3.44	1.63	36
Relaxing / calming visuals	3.35	1.60	37
Humor / lighthearted content	3.35	1.51	37
Collection (achievements, rewards)	3.22	1.59	36
Creativity	3.17	1.34	36
Cooperative play	3.08	1.46	36
Problem-solving	2.86	1.67	37
Physical activity (movement-based)	2.14	1.67	37

Grouping by conceptual family reinforces the mechanic-level pattern. Narrative-oriented features (stories and exploration) produced the highest grouped mean ($M = 3.51$); aesthetic and tonal features (relaxing visuals, humor) followed ($M = 3.35$); social features (cooperative play, creativity) fell in the moderate range ($M = 3.13$). Achievement and progress features were somewhat lower ($M = 3.04$), and physical engagement was lowest ($M = 2.14$). Students in stress contexts consistently favored low-demand, interpretive, and narrative features over active or performative ones.

5.2.4 Progress Tracking Preferences

A third item set asked how meaningful four progress-tracking formats were for supporting wellness goals (1 = *not at all important*, 5 = *extremely important*; Table 5.3). All four fell below the midpoint. The narrative journey map attracted the highest

endorsement ($M = 2.89$), while mood charts scored lowest ($M = 2.11$). Streak calendars showed the highest variability ($SD = 1.93$), indicating polarized reactions: some students found them motivating; others associated them with guilt around missed days.

Table 5.3: Ratings of progress-tracking mechanics for wellness goal support (1 = *not at all important*, 5 = *extremely important*).

Progress-Tracking Mechanic	M	SD	N
Journey or quest map reflecting achievements	2.89	1.83	36
Progress toward personal wellness goals	2.72	1.83	36
Streak / consistent days calendar	2.57	1.93	35
Weekly or monthly mood chart	2.11	1.45	36

The fact that even the best-rated format — a narrative journey map — fell below the midpoint indicates that progress tracking as a category is less valued by students than narrative engagement or personalization. This finding directly shaped the decision to frame advancement in *The Archive* as narrative movement rather than metric accumulation.

5.2.5 Stress Sources and Coping Activities

Academic pressure was rated most frequently as a stressor ($M = 3.57$, $SD = 1.07$), followed by time management ($M = 2.92$) and financial concerns ($M = 2.81$; Table 5.4). Among coping activities (Table 5.5), talking to friends and family received the highest usefulness rating ($M = 3.70$), followed by taking breaks ($M = 3.35$). Using mental health applications received the lowest rating of all eight options ($M = 1.03$, $SD = 1.38$), far below the scale midpoint. This was not a reflection of non-exposure: 14 participants reported actively using such tools. The low endorsement situates gamification preferences within a broader context of dissatisfaction — students did not simply prefer certain mechanics, they found the context in which those mechanics appeared broadly ineffective.

Table 5.4: Frequency ratings for primary stress sources (1 = *never*, 5 = *always*).

Stressor	<i>M</i>	<i>SD</i>	<i>N</i>
Academic pressure	3.57	1.07	37
Time management	2.92	1.36	37
Financial concerns	2.81	1.61	37
Loneliness or isolation	2.33	1.47	27
Family expectations	2.21	1.89	24
Social life and relationships	2.03	1.36	35
Physical health	1.48	1.22	27

Table 5.5: Perceived usefulness of coping activities (1 = *not at all useful*, 5 = *extremely useful*).

Coping Activity	<i>M</i>	<i>SD</i>	<i>N</i>
Talking to friends or family	3.70	1.51	37
Taking breaks or vacations	3.35	1.55	37
Physical exercise or sports	2.73	1.73	37
Setting personal goals	2.27	1.63	37
Journaling or self-reflection	1.89	1.51	36
Counseling or therapy	1.75	1.66	36
Meditation or mindfulness	1.61	1.36	36
Using mental health apps	1.03	1.38	35

Participants were also moderately open to team-based self-care challenges ($M = 3.59$) but showed little interest in receiving group-based motivational messages ($M = 1.95$), reinforcing the preference for peer connection without public performance pressure.

Taken together, the Phase I data produced a clear student preference profile: narrative and personalization features were valued; competitive, evaluative, and metric-tracking features were not. This profile directly shaped three design principles carried into *The Archive*: narrative exploration over task completion, avoidance of leaderboards and streaks, and personalization as a structural value rather than a cosmetic one. With this quantitative baseline established, the Phase II focus groups explored *how* students experience and narrate stress — the qualitative ground that the design would need to stand on.

5.3 Phase II Findings: Narrative Focus Groups

Eleven focus groups with $N = 30$ undergraduate and graduate students produced six themes organized into two categories: how narrative roleplay functioned as a space for emotional and social sensemaking, and how the roleplay experience shaped participants' expectations for digital wellness mechanics. Each participant is cited by group-participant code (G#-P#).

5.3.1 Theme 1: Roleplay as a Reflective Scaffold

Participants used the fictional setting of Blackthorn University to explore emotions, test interpersonal strategies, and reflect on stress responses in ways that direct interview would not have supported. The psychological distance of character enactment allowed students to articulate experiences of fear, failure, and relational burden more openly. In Group 2, a participant playing the “Shadowed One” offered an unprompted in-character monologue:

“In the guise of the Shadowed One, I could say my darkest fears out loud without feeling judged. I have been hated. I have been hunted, mistrusted. The only person who believed in me, I let down.” (G2-P1)

During debriefs, participants frequently described alignment between their character's choices and their own real-world coping tendencies. One participant in Group 1 noted: *“I’m very afraid of putting weight on other people. Whereas the character... that would probably be more of like a moment of self-reflection for me”* (G1-P1). Others used characters to practice coping approaches they would not typically employ. As one participant in Group 6 put it: *“I liked being able to mess up or do something weird and know it was just part of the character”* (G6-P3). Avatars progressively functioned as mirrors rather than masks — shared reference points through which participants externalized stress and rehearsed responses in a setting where social risk was contained.

5.3.2 Theme 2: Trust, Conflict, and Collaborative Repair

Narrative prompts that invited blame assignment and alliance formation surfaced interpersonal dynamics that mirrored real-world stress behaviors: scapegoating, avoidance,

and social withdrawal. In Group 3, a participant described the experience of entering the scenario already under suspicion: *“They immediately looked at me like I was guilty. I had not even said anything yet. It made me feel like I had to prove my innocence from the start”* (G3-P1).

By Act Two, most groups transitioned from accusation toward cooperative problem-solving. A participant in Group 5 reflected: *“It felt like we were in a group project with real stakes. I wanted to fix things with them, not win”* (G5-P1). In Group 3, participants built a shared memory map following a betrayal scene to restore cohesion. One participant recalled: *“It felt like we had gone through something together. Even though we made it up, it mattered”* (G3-P2). These acts of collaborative repair — using collective decision-making to re-establish trust and shared purpose — were among the most consistent dynamics across all eleven groups and informed the design of The Archive’s Curator as a guide who explicitly supports rather than evaluates the player.

5.3.3 Theme 3: Preference for Flexible, Peer-Anchored Mechanics

Across all eleven groups, participants critiqued rigid gamification elements as structurally misaligned with how academic stress actually operates. Static progress bars were described as motivating only until completed: *“Once you fill up this bar, there is no incentive to do it again. It just sits there”* (G5-P3). Leaderboards drew more pointed criticism, redirecting attention from the wellness goal toward performance ranking: *“If I am trying to improve my habits, I would say no to leaderboards. I tie too much meaning to where I rank instead of focusing on the actual habit”* (G5-P2).

In their place, participants described flexible systems that could adapt to shifting emotional states: *“If I am not in the mood for one kind of challenge, I should be able to switch it up. Mental health is not one-size-fits-all”* (G2-P2). Peer support was valued under conditions that kept it ambient and non-performative — awareness of others’ engagement without direct comparison or required interaction.

5.3.4 Theme 4: Preference for Story-Based Goal Setting

Participants consistently preferred story-driven goal structures over checklists, badges, and habit trackers. The appeal was not tied to any specific plot but to shared narrative

properties: meaningful choice, unfolding consequences, and emotional investment in outcomes. One participant captured the motivational logic directly:

“If I had a couple different options and I were reasonably invested in the mini story going on, then I would say, oh, that’s the one I want to do. If I want to do it, I’m going to naturally want to do the tasks associated with it.” (G2-P2)

Narrative pacing sustained engagement in ways that static mechanics did not: *“When the story builds up and you know something big is coming, it keeps you engaged. You want to see what happens next”* (G3-P1). Participants in the co-design phase sketched progress-tracking interfaces organized around story movement rather than percentage-complete indicators, directly informing *The Archive*’s narrative advancement structure.

5.3.5 Theme 5: Mood Tracking and Reflection Require Redesign

Participants across groups criticized existing mood logs for over-representing distress and under-representing recovery or positive states. One participant described the practical consequence: *“I saw that I mostly inputted my negative things and I was feeling good or happy. I didn’t ever change that”* (G1-P2). Several participants proposed passive tracking as a lower-burden alternative:

“What if it was just tracking how active I was and what time I’m up and moving and kind of giving me like a graph and then it would put in like an emoji... this is what you felt based on your patterns?” (G1-P1)

Others called for reflection that went beyond labeling: *“Ask what led up to the emotion. Ask what I did to handle it. That’s more helpful than just picking a face or color”* (G2-P2), and for feedback that acknowledged positive states alongside difficult ones: *“It should show when things are going good too. Like a little high five or something”* (G2-P2). These observations shaped the Codex journaling system in *The Archive*, which offers optional, variable-depth prompts across text, drawing, and voice modalities rather than daily emotional logging.

5.3.6 Theme 6: Wellness Design Should Reduce Friction and Avoid Overwhelm

The most consistent finding across all eleven groups was that a wellness tool must not become another obligation. Extensive setup, frequent notifications, and wearable hardware requirements were described as emotionally exhausting: *“I don’t want this to feel like a chore for anybody”* (G1-P3). Notification systems were specifically named as a source of anxiety rather than motivation: *“Is this something extra I’m going to have to think about during the day? And more specifically, is this something where I’m going to feel bad if I forget?”* (G1-P2). Participants’ preferred entry conditions were immediate and frictionless: *“Ideally you want the barrier to entry to be as low as possible so that you can just download the app and get started. Make it feel effortless”* (G1-P1).

These preferences were operationalized in *The Archive* through a brief onboarding sequence, no mandatory check-ins or notifications, sessions completable in a single low-effort pass, and all reflective activities framed as optional rather than required for narrative progression.

5.3.7 Summary: From Focus Groups to Design

The six themes collectively established the experiential and motivational landscape that *The Archive*’s design needed to address. Stress was revealed as cyclical, relational, and interpretive — not a fixed state amenable to metric tracking. The consistent student design vocabulary was: flexible over rigid, story-driven over checklist-driven, ambient over performative, and low-friction over feature-dense. With this participatory foundation in hand, the design was built and subsequently submitted to expert evaluation before deployment.

5.4 Expert Evaluation Findings

Ten domain experts participated in individual think-aloud sessions with *The Archive* prior to Phase III deployment: two evaluators brought expertise in human–computer interaction and game design (E2, E3), one in creative writing and narrative craft (E3),

one in education and personality assessment (E4), one in philosophy of play and serious games (E5), and one in mental health counseling (E6). Two evaluators (E2, E3) completed a structured quickfire rating protocol on a 1–5 scale at session’s end; the remaining four sessions were qualitative. Table 5.6 presents the structured ratings; cross-session thematic analysis identified eight themes reported below with observations cited by evaluator code.

Table 5.6: Quickfire Likert ratings from evaluators who completed the structured protocol (E2, E3). Scale: 1 = not at all / strongly disagree; 5 = completely / strongly agree.

Dimension	E2	E3
Interface clarity and navigation ease	4	2
Choices felt meaningful and self-directed	4	3
Feedback supported understanding/progression	3	3
Character/story fostered emotional connection	4	2
Visuals/pacing supported absorption	3–4	2
Experience felt engaging and motivating	4	3
Stress themes handled sensitively	3–4	4
Story encouraged self-awareness/insight	4	2–3

5.4.1 Onboarding and Narrative Grounding

Across all six sessions, evaluators encountered a gap between the opening screen and a felt sense of what the experience was for. The landing page presented two onboarding pathways without first orienting users to the narrative context. E4 articulated the issue: *“It looks like there’s an assumption that when you come here, this is your starting point, do this or do this. But in terms of what is this all about? Why am I getting here?”* Their recommendation was an inviting welcome message that positioned the experience as an embarkation: *“What would have been helpful would be something like, come along here and embark on this part with me.”* E3 found the tutorial information useful but its sequential front-loaded delivery — all blocks presented before first interaction — inconsistent with how games typically introduce their worlds. E1 noted that the tutorial’s tone effectively communicated that players did not need to remember everything, a successful design choice whose benefit was undermined by the absence of a clear path

from tutorial to first story.

5.4.2 Interface Readability and Accessibility

E3, who assigned a rating of 2 out of 5 for interface clarity, identified text size, passage length, and foreground–background contrast as insufficient: *“I was able to navigate it, but it took me a while sometimes to understand what I was reading, either because of length, size of text, or color combinations.”* The evaluator noted accessibility concerns for users with visual impairments. Multiple evaluators independently flagged the absence of visual differentiation between narrative prose and dialogue choices — a distinction that readers in text-based experiences rely on formatting conventions to make. E3 suggested italics or color shifts; E5 independently encountered a practical low-contrast failure when an exit button appeared blue on blue. Long blocks of yellow text on a dark background were noted by several evaluators as visually demanding; E3 suggested that even a static scene-relevant background image would substantially improve perceived readability.

5.4.3 Player Agency and the Curator’s Tone

Agency generated the most sustained discussion across sessions. Evaluators consistently valued the choice structure but identified two patterns that attenuated it: non-consequential choice points and a Curator dialogue style that risked reading as dismissive. E3 (autonomy rating: 3/5) identified the locked-door sequence as the clearest instance of a non-advancing choice: *“Why am I checking doors if they’re locked? Another way to frame that is like improv storytelling: if the door is locked, is there a yes-and component?”* On Curator tone, the same evaluator gave a precise analysis: *“The tone was dismissive. I don’t necessarily think that is the case. I think it’s a simple matter of without a visual or any other thing, there’s a lot more room for me to infer a dismissive tone than there actually is.”* The clinical implication was also raised: *“The curator has specified themselves as an ally. So even though there’s an unexplicitly dismissive tone, that disconnecting feeling would be like, Oh, here’s a wall. I’m not connecting.”* E5 offered a structural suggestion: keep the Curator’s framing grounded in narrative language rather than surfacing external research rationale through the character.

5.4.4 Narrative Immersion and Visual Design

Evaluator responses to the narrative text were broadly positive in tone while identifying the gap between evocative description and the experience of reading it without visual support. E3 (relatedness and presence: both 2/5) articulated the issue: *“There are moments where it’ll say you’re seeing this, you’re sensing this, but obviously, on my end, it’s just a black screen with some yellow text.”* The evaluator estimated that even a static scene image would increase immersiveness substantially. By contrast, E6 — a mental health counselor with no gaming background — experienced the opening sequence as immediately calming and immersive, reading it as a guided visualization: *“I felt like I was reading it in a very soothing voice. It had a calming feel and definitely some curiosity about where I was headed.”* This divergence is interpretively significant: the text quality was sufficient for non-gamers without comparative frameworks for interactive narrative, while experienced players used that framework to identify gaps between the stated and the rendered experience. E2 compared the application favorably to *Celeste*, citing it twice as a positive benchmark. E1 identified the dark-room entry sequence — offering the player choices about how to orient in an unknown space — as the moment of strongest emotional presence.

5.4.5 Reflective Journaling: Integration and Design

All six evaluators found the multimodal journaling design — text, drawing, and voice — meaningful. The consistent concern was not the journaling itself but its felt disconnection from the narrative. E2 (reflection: 4/5) described it as a *“valuable part”* while noting that prompts drawn from general stress research felt more separate from the story than a prompt tied to a specific narrative moment would. E5 confirmed the distinction: the prompt asking for a quality the Curator had that the evaluator wished others possessed felt narrative-grounded and earned; the other prompts felt detached. E4 raised a structural suggestion: rather than three separate prompts with separate modalities, present one prompt with a choice of response format. E6 recommended validation as the most appropriate journaling reward — a brief affirming message rather than an attribute gain — reflecting the therapeutic logic of reflection as a practice rather than a performance.

5.4.6 Mind Defender: Concept and Execution

The Space Invaders-style thought reframing mini-game received consistent conceptual endorsement alongside consistent execution concerns. E6 described thought reframing as clinically resonant: *“That is something that I often teach — reframing. We can just get stuck with negative thinking and we need to rewire our brains to look at things differently.”* E3 called the concept *“fantastic.”* The execution challenges clustered around three issues: thought-bubble text was too small and fast to read at gameplay speed, the collect/shoot mechanic was not initially legible (E2 spent several rounds avoiding green bubbles rather than collecting them), and the paddle control zone was not self-evident as the primary interaction surface. Suggested improvements included simpler single-word thought labels (E5), a post-game summary of thoughts encountered (E4), integration of the player avatar as the controlled element (E3), and in-story narrative justification for the activity through Curator dialogue (E3, E5).

5.4.7 Game Loop Coherence and Reward Transparency

Evaluators found the three-part loop — story, journaling, mini-game — structurally coherent but the connections between components insufficiently visible. E2 described the activities as individually valuable but experientially siloed. E3 framed the underlying problem: *“The causality between one and the other is going to make all of it feel more cohesive and vital. Everything that goes into that shopping cart checks out at the end.”* On rewards, E4 described the attribute gain notification as something they did not initially recognize as a reward; E1 asked whether a “fear plus one” gain was good or bad, unable to determine the valence from the label alone. Both recommended brief contextual glosses at the moment of receipt. E5 raised a structural concern: reward metrics visible to the player should live within the narrative world rather than appearing as external research constructs, trusting that the story itself provides sufficient justification for engagement.

5.4.8 Therapeutic Alignment and Emotional Safety

E6 provided the most direct therapeutic evaluation. The application was described as “*gentler*” than direct intervention — a complement to counseling that might reach students who would never seek clinical support, with an indirect metaphorical approach that reduced help-seeking stigma. The thought reframing exercise was clinically validated as a standard technique. E6 recommended neutral emotional language in early story prompts (e.g., “*you find yourself curious about*” rather than “*you are scared*”) to avoid prescribing states that might not match a user’s experience. On the question of emotional depth, E6 supported introducing darker themes — including nihilism and the feeling of being trapped — while recommending that choices within those themes remain emotionally open so players could select responses accurate to their own position. E5 offered the most expansive framing of the application’s potential: “*The interesting original thing you’re doing is really leaning into the value of games themselves and letting that be valuable for mental health. Other people have already done the gamified mental health tidbits stuff.*” The evaluator recommended positioning the work firmly as a serious game, trusting that narrative and play would do the reflective work without requiring external justification surfaced to the player.

5.4.9 Revisions Prior to Phase III

The six sessions informed five targeted pre-deployment revisions: (1) text contrast and font size improved across all passage screens; (2) Curator dialogue revised at redirection junctions to clarify supportive intent without altering the character’s voice; (3) a brief welcoming narrative message added to the onboarding screen before the pathway choice; (4) optional framing of journaling and mini-game made explicit in tutorial language; and (5) attribute reward notifications given one-sentence narrative-grounded glosses at the point of receipt. With these revisions incorporated, the application proceeded to Phase III deployment.

5.5 Pre/Post Wellbeing Outcomes

Fifteen participants were matched across pre- and post-surveys. Three who enrolled on April 2, 2026 completed only the pre-survey and are excluded from change score

analyses. One test response was removed. Change scores are calculated as $\Delta = \text{Post} - \text{Pre}$; negative PSS change indicates stress reduction. Table 5.7 summarizes the matched sample.

Table 5.7: Pre/Post Summary Statistics ($N = 15$ Matched Participants)

Measure	Pre M	Post M	Mean Δ	SD of Δ	Improved	Worsened
PSS (0–40)	21.40	18.93	−2.47	4.69	9/15 (60%)	5/15 (33%)
Flourishing (8–56)	43.27	45.20	+1.93	2.05	11/15 (73%)	1/15 (7%)
HEMA-r (1–7)	4.74	5.10	+0.36	0.52	10/15 (67%)	3/15 (20%)

5.5.1 Perceived Stress

Average PSS decreased by 2.47 points, from 21.40 to 18.93. Both means fall within the moderate stress range (14–26) (**cohen1983**); the group moved toward the lower end of that range following engagement. Nine of fifteen participants (60%) showed stress reduction; one was unchanged; five showed modest increases (range: +1 to +3 points).

Individual variation was substantial ($SD_{\Delta} = 4.69$). Three participants showed clinically meaningful improvements: P44 (SI2954) dropped 12 points from the high into the moderate range; P49 (AB9703) dropped 10 points from moderate into the low range; P47 (ST7149) dropped 8 points, also reaching the low stress range by post-survey. The five participants whose PSS increased showed changes of 1–3 points, unlikely to represent meaningful change given measurement sensitivity and the absence of a control condition. These mixed patterns are consistent with an intervention that benefits a meaningful proportion of participants without uniform effect — an expected finding for any wellness system without strong behavioral prescriptions.

5.5.2 Flourishing

Flourishing showed the most consistent movement across the three measures. Eleven of fifteen participants (73%) improved; three were unchanged; only one showed any decline — a single-point decrease from 48 to 47. The SD of change scores (2.05) was notably smaller than that of PSS (4.69), indicating more consistent directional movement despite variation in magnitude. The group mean rose from 43.27 to 45.20, with both means

well above the scale midpoint, suggesting possible ceiling effects for participants who began with high wellbeing. Participants with lower baseline scores showed larger gains, consistent with greater room for improvement.

The directionality is particularly notable for an intervention of this duration. Five days of engagement with a narrative application would not typically be expected to produce detectable group-level Flourishing change. That 73% of participants improved while only 7% showed any decline — and that decline was a single point — suggests the experience supported something meaningful in participants’ sense of engagement, meaning, or positive self-perception. This interpretation is strengthened by the significant correlation between PLAY Heuristics Narrative scores and Flourishing reported in Section 5.8: participants who rated the narrative quality more highly also reported higher wellbeing, suggesting that narrative coherence was a plausible mechanism.

5.5.3 Motivation to Engage

HEMA-r scores increased from 4.74 to 5.10, a change of 0.36 points ($SD = 0.52$). Ten of fifteen participants (67%) showed increased motivation; three showed decreases; two were unchanged. While the smallest of the three effects, the direction is consistent with a design that supports intrinsic engagement rather than producing the guilt and avoidance documented in compliance-oriented systems ([hoffmann‘gamification’2017](#)).

5.6 Daily Ecological Momentary Assessment

5.6.1 Compliance and Daily Trends

Forty-seven EMA responses were collected from 18 users across five days, yielding an overall compliance rate of 52.2% (maximum possible: 90). Daily response counts ranged from 7 (Day 2) to 14 (Day 5; 78% compliance). The high Day 5 compliance suggests that participants who remained engaged found the experience valuable enough to continue through the final module.

As shown in Table 5.8, the three EMA measures diverged across the week: in-the-moment stress decreased while coping confidence and perceived worth of engagement increased.

Table 5.8: Daily EMA Means by Study Day

Day	N	Stress (SD)	Worth It	Coping	Compliance
Day 1	8	4.25 (3.49)	3.75	5.50	44%
Day 2	7	4.14 (2.54)	3.71	6.71	39%
Day 3	7	2.86 (2.48)	3.43	5.00	39%
Day 4	11	3.09 (2.47)	3.55	5.82	61%
Day 5	14	2.43 (2.53)	4.21	7.00	78%
Overall	47	3.21 (2.68)	3.79	6.13	52%

In-the-moment stress declined 1.82 points from Day 1 to Day 5, mirroring the direction of pre/post PSS change and providing convergent within-week evidence that stress was moving positively across the study period. Day 3 stands out: stress dropped while coping confidence also dipped to its lowest point (5.00), suggesting that the Poppy module — which asks players to identify what stings most about their own stress — was emotionally confronting without immediately increasing coping confidence. The dip recovered by Day 4 and rose substantially by Day 5, consistent with a narrative arc in which emotionally demanding content precedes resolution.

Coping confidence showed the largest proportional gain (+1.50 points, Day 1 to Day 5), providing direct within-session evidence that the narrative and reflective activities were associated with increased perceived coping capability in the moment. Worth It increased from 3.75 to 4.21 — suggesting participants found the experience increasingly valuable as the week progressed, the opposite of the declining engagement associated with compliance-oriented systems (hoffmann'gamification'2017).

5.6.2 Individual Stress Trajectories

Nine participants with three or more EMA responses could be assessed for within-person trajectories. Table 5.9 presents these patterns. Six of nine (67%) showed net improvement; two were stable; one showed a 1-point increase.

Table 5.9: Individual EMA Stress Trajectories ($N = 9$ with 3+ responses)

User	Stress Trajectory	Direction	Δ	Coping Trend
U45	D1:6→D3:2→D4:4→D5:6	Stable	+0	Fluctuating
U46	D2:2→D4:0→D5:0	Improved	−2	High, stable
U47	D2:3→D3:1→D4:0→D5:1	Improved	−2	Rising
U48	D1:2→D3:8→D4:7→D5:0	Improved	−2	Rising
U49	D1:0→D3:4→D4:0→D5:1	Worsened	+1	Unstable
U50	D1:1→D3:1→D4:2→D5:0	Improved	−1	Rising
U54	D1:4→D2:8→D4:2	Improved	−2	Rising
U56	D3:2→D4:4→D5:1	Improved	−1	Rising
U57	D2:3→D4:4→D5:3	Stable	0	Slightly declining

Three trajectories merit specific attention. U48 spiked sharply to 8 on Day 3 before dropping to 0 on Day 5 — an example of temporary destabilization followed by stabilization, consistent with how narrative-based wellness might work when it is asking participants to engage with difficult material rather than avoid it. Notably, coping confidence rose across the same arc (1→2→5→6). U47 showed the clearest steady decline and, across all available data sources (pre/post PSS, EMA stress, and EMA coping), presented a consistent picture of improvement. U45’s unchanged trajectory from first to last response — stress returned to its Day 1 level — is the most ambiguous: whether this reflects disengagement with the narrative, external life circumstances, or simple non-response would require qualitative follow-up to determine.

5.6.3 EMA Internal Correlations

Pearson correlations across 47 EMA responses (critical $r \approx 0.288$ for $p < 0.05$, two-tailed) revealed one strong significant relationship (Table 5.10).

Table 5.10: EMA Internal Correlations ($N = 47$ responses)

Variable 1	Variable 2	r	Sig.
Stress Now	Worth It	−0.18	No
Stress Now	Coping Confidence	−0.16	No
Worth It	Coping Confidence	0.767	Yes*

* $p < 0.05$

Worth It and Coping Confidence were strongly correlated ($r = 0.767$): participants who found engagement worthwhile in a given moment also reported substantially higher coping confidence in that same moment. This association holds a direct interpretive implication: the mechanisms that made the experience feel worth engaging with — narrative quality, meaningful choice, emotional resonance — may be the same mechanisms producing the coping confidence gains observed across the week. The non-significant and near-zero relationships between in-the-moment stress and both other measures are equally important: how stressed a participant was during engagement did not predict whether they found it worthwhile or felt capable of coping. The experience was accessible and felt valuable across a range of stress states.

5.7 Integrated EMA and Survey Analysis

EMA user identifiers were linked to pre/post participant numbers for the 15 matched participants. For $N = 15$, the critical $r \approx 0.514$ ($p < 0.05$, two-tailed). Table 5.11 presents significant integrated correlations.

Table 5.11: Significant EMA–Survey Integrated Correlations ($N = 15$)

EMA Measure	Survey Outcome	r	Interpretation
Avg EMA Stress	Post PSS	0.734	Convergent validity
Day 5 EMA Stress	Post PSS	0.622	Proximal single-day validity
Avg EMA Worth It	ETES Overall	0.847	Strongest integrated finding
Avg EMA Worth It	GAMEX Agency	0.517	Worth It linked to agency
Avg EMA Coping	ETES Overall	0.691	Real-time coping predicts transformation

Average EMA Stress correlated significantly with post-survey PSS ($r = 0.734$), confirming that the brief embedded stress item tracked the same construct as the validated PSS-10. This convergent validity is encouraging: the EMA data can be interpreted alongside the survey data with confidence. Importantly, PSS change scores were not predicted by average EMA stress ($r = 0.122$, non-significant), indicating that the EMA tracked participants’ general stress level rather than the direction of change.

The strongest correlation in the entire dataset was between average EMA Worth It

and ETES Overall ($r = 0.847$). Participants who consistently rated their engagement as worthwhile reported significantly higher emotional transformation and eudaimonic experience quality at post-survey. This finding is practically meaningful: the ETES assesses growth, self-actualization, and authentic self-expression — qualities that narrative wellness systems aim to support — and these were best predicted not by any specific rated design element but by the participants’ own in-the-moment sense that the experience was worth their time. Average EMA Worth It also correlated with GAMEX Agency ($r = 0.517$), linking momentary perceived worth to retrospective sense of meaningful control.

Average EMA Coping Confidence correlated with ETES Overall ($r = 0.691$): in-the-moment coping self-efficacy during narrative engagement predicted post-survey emotional transformation, providing evidence that the system worked through a coping-enhancement pathway rather than solely through entertainment or distraction.

5.8 Post-Survey Correlations

5.8.1 Post-Survey Descriptive Statistics

Table 5.12 presents descriptive statistics for all post-survey scales ($N = 15$). The post-survey PSS mean (18.93) remained in the moderate stress range. The Flourishing mean (45.20 out of 56) indicates relatively high psychological wellbeing. GAME-X, PLAY Heuristics, and ETES scores ranged from 3.55 to 3.97 on 1–5 scales; the ETES showed the most variance ($SD = 0.67$), consistent with meaningful individual differences in how transformative participants found the experience.

Table 5.12: Post-Survey Descriptive Statistics ($N = 15$)

Scale	<i>M</i>	<i>SD</i>	Min	Max	Range
PSS (Post)	18.93	5.15	10	27	0–40
Flourishing (Post)	45.20	3.78	35	49	8–56
HEMA-r Overall	5.10	0.63	–	–	1–7
GAME-X Overall	3.63	0.48	–	–	1–5
PLAY Heuristics Overall	3.97	0.60	–	–	1–5
ETES Overall	3.55	0.67	–	–	1–5

5.8.2 RQ1: Design Quality and Wellbeing Outcomes

No significant relationships were found between post-only game experience scales and post-survey PSS. This is expected in a cross-sectional post-only design where PSS variance reflects pre-existing individual differences rather than intervention effects. Two significant Flourishing correlations bore on RQ1. PLAY Heuristics Overall correlated with Flourishing ($r = 0.507$, $p < 0.05$): participants who rated overall playability more highly reported higher psychological wellbeing. HEMA-r Hedonic motivation also correlated with Flourishing ($r = 0.601$, $p < 0.05$): participants who entered seeking enjoyment and relaxation reported higher flourishing, suggesting that hedonic orientation may have facilitated more open, receptive engagement with the narrative.

5.8.3 RQ2: Design Elements, Agency, and Transformation

The analysis bearing most directly on RQ2 identified a cluster of significant relationships centered on agency, emotional resonance, and transformative experience (Table 5.13).

Table 5.13: Significant Post-Survey Correlations: RQ2 ($N = 15$, $p < 0.05$)

X	Y	<i>r</i>	Interpretation
GAMEX Agency	ETES Authenticity	0.842	Agency predicts authentic self-expression
PLAY Emotion	ETES Overall	0.781	Emotional resonance drives transformation
GAMEX Agency	PLAY Emotion	0.754	Agency enables emotional connection
PLAY Narrative	Flourishing	0.568	Narrative quality supports wellbeing

The strongest correlation in the post-survey dataset was between GAMEX Agency and ETES Authenticity ($r = 0.842$). Participants who felt meaningful control during gameplay — genuine choice rather than cosmetic variation — were dramatically more likely to report authentic self-expression through the experience. GAMEX Agency also correlated with PLAY Emotion ($r = 0.754$): participants who felt in control also felt more emotionally connected to the narrative, indicating that agency and emotional engagement were linked rather than independent. PLAY Emotion in turn correlated with ETES Overall ($r = 0.781$), suggesting a plausible chain: agency enables emotional connection; emotional connection predicts transformation. PLAY Narrative correlated

with Flourishing ($r = 0.568$), linking the coherence and depth of narrative structure to general psychological wellbeing.

PLAY Engagement showed a small negative trend with PSS ($r = -0.241$, n.s.), suggesting that engagement time or immersion alone is not the active ingredient in stress reduction. The broader pattern consistently points toward agency, emotional resonance, and perceived worth as the central mechanisms of effect.

5.8.4 RQ3: Motivational Alignment and Participatory Design

RQ3 is addressed at the quantitative level by examining whether participants whose motivational profiles matched the design’s values showed stronger outcomes. Table 5.14 presents the relevant significant correlations.

Table 5.14: Significant Post-Survey Correlations: RQ3 ($N = 15$, $p < 0.05$)

X	Y	<i>r</i>	Interpretation
Eudaimonic (HEMA-r)	ETES Self-Actualization	0.727	Growth motivation predicts self-actualization
Eudaimonic (HEMA-r)	ETES Overall	0.718	Growth motivation predicts transformation
Eudaimonic (HEMA-r)	GAMEX Agency	0.633	Growth motivation enables agency
HEMA-r Overall	GAMEX Overall	0.520	Motivation predicts game experience

Eudaimonic motivation was the most consistent predictor across multiple outcomes. Participants seeking growth, meaning, and personal development were significantly more likely to report self-actualization ($r = 0.727$), overall transformative experience ($r = 0.718$), and a stronger sense of agency ($r = 0.633$). HEMA-r Overall also predicted GAME-X Overall ($r = 0.520$). These findings are consistent with the interpretation that the participatory design process successfully aligned *The Archive* with students’ intrinsic motivational orientations: participants who already sought the qualities the design embedded were most likely to find them. The caveat is that dispositionally eudaimonic participants may find any reflective experience more transformative; a comparative design would be required to disentangle design-specific fit from dispositional tendency.

5.9 Phase III Qualitative Findings: Exit Interview Analysis

Exit interviews were conducted with all 15 participants (P43–P58) following the five-day deployment period. Interviews were audio-recorded and transcribed. Analysis followed Braun and Clarke’s thematic analysis framework (**braun’using’2006**), proceeding from open coding of participant language through iterative refinement across the full sample. Five themes emerged. Each is reported below with representative participant quotations cited by participant code.

5.9.1 Story–Experience Alignment

Participants consistently recognized the narrative themes as reflective of their own stress experiences. This recognition occurred even among participants who did not report acute stress during the deployment week. Imposter syndrome framing in the Embark module and the peer dynamic in the Poppy module were the most frequently cited points of alignment.

P44 described identifying with the final chapter’s framing of authenticity under social pressure: *“I realized, oh yeah, that’s true. Maybe what I have done throughout the story is what being true to myself, instead of like living on other people’s thoughts.”* P46 described a similar moment of recognition in the Poppy module: *“When Poppy was describing what the experience was exactly about, I felt like, yeah, I actually feel that particular feeling sometimes in my daily life.”*

Participants who reported lower stress during the week still recognized the design model as applicable. P56 articulated the value of the Archive framing in general terms: *“The idea of this thing that is always available and has a high likelihood of being returned to — I think that’s very helpful to remember as maybe large-scale mental health.”* This pattern suggests that story–experience alignment operated at the level of recognition rather than requiring contemporaneous distress.

5.9.2 The Archive as Low-Stakes Behavioral Rehearsal

Several participants described using the choice system to explore responses they would not enact in real situations. P43 observed that in-story choices tracked their real-time

stress state: *“The days I was more stressed, I was going for the safer answers. And on the days I was feeling more playful, I was going for the playful options.”* P44 articulated the rehearsal function more directly: *“Since it’s a game, they’re truly expressing their feelings and getting to know themselves, what’s going on in their life. In a real situation, they might have considered other people’s thinking.”*

P51 described the avatar structure as providing sufficient psychological distance to enable genuine self-reflection: *“It kind of made it feel like I was taking a step away from myself and it was actually just acting as the avatar in the game. And through that I was able to actually start doing some more thinking about self stuff.”* This distancing effect appears to lower the threshold for engagement with stress-related content, functioning as a form of behavioral rehearsal within a protected narrative frame.

5.9.3 Non-Punitive Engagement Model

Participants responded positively to the absence of failure states and performance metrics. P47 noted that the application produced a relaxation effect distinguishable from other games they engaged with regularly: *“I play video games pretty much every day, but I think this is one of the very few where playing it actually did some sort of relaxation-type effect.”* P55 connected the experience to self-compassion: *“Maybe it could be good to feel better about giving myself grace in terms of my stress, allowing myself those little bits.”*

P56 articulated the non-punitive logic most precisely, describing the application as a tool oriented toward availability rather than obligation: *“It does not need to be a consistent thing. It does not need to be a set schedule thing, but something that is always available. And I think that’s very helpful to remember.”* This framing is consistent with Phase II focus group feedback, in which participants rejected streak-based mechanics and described existing wellness applications as obligations. The Phase III data suggest that participants recognized and internalized the intended design value.

5.9.4 The Curator as Guide Character

Participant responses to the Curator were generally positive, with notable variation in legibility. P51 described the narrative reveal in the final module as reorienting: *“At*

the end-ish, when it was revealed like, I'm just a reflection of you — that really kind of nailed in the themes. It kind of reoriented my own opinions a little bit more." P44 reported experiencing the Curator's framing as therapeutically significant.

Several participants, however, indicated uncertainty about the Curator's identity and role within the world. P47 and P56 both noted they were unclear on who the Curator represented or why he was present, suggesting that the character's contextualizing function was not consistently legible without additional scaffolding. This finding is consistent with expert evaluator feedback from §5.4 that recommended clearer onboarding for the narrative frame.

5.9.5 First Chapter and Narrative Entry Experience

Most participants reported navigating the Embark module without significant difficulty. The withholding of explanatory context in favor of environmental immersion was generally accepted. P45 described this orientation: *"It definitely gets into some heavy topics, but I think it does it in a good way where it gives you tools for dealing with it, albeit in a bit of an abstract way."*

The primary exception was P58, who reported a more significant entry barrier and did not complete all modules during the deployment window. P53 reported a related finding: the application functioned as an effective distraction but did not engage with their specific academic stress context, which centered on examination pressure that the narrative metaphors did not directly address. These cases identify specific design targets for future iteration, particularly for participants whose stress experiences are domain-specific rather than diffuse.

5.9.6 Relationship to Quantitative Findings

The five qualitative themes triangulate with the quantitative findings reported in §5.5–5.8. The consistent participant endorsement of the non-punitive model and the behavioral rehearsal function across varied stress levels supports the interpretation that the design values developed through Phases I and II translated into the deployed experience. The significant correlation between GAMEX Agency and ETES Authenticity ($r = 0.842$; Table 5.13) is consistent with participants' accounts of the choice system as

enabling genuine self-expression. Variability in Curator legibility and first-chapter navigation — where difficulty was the exception rather than the norm — identifies specific components for future design iteration.

5.10 Phase II to Phase III Design Traceability

Table 5.15 traces the thread from Phase II focus group themes through specific design decisions in *The Archive* to Phase III evidence bearing on each decision. This traceability analysis addresses RQ3 by documenting the provenance of each major design choice and the degree to which deployment findings validated or complicated it.

Table 5.15: Phase II Theme to Design Decision to Phase III Evidence

Phase II Theme	Design Decision	System Element	Phase III Evidence
Preference for story-driven goal structures over checklists and badges (Theme 4; G2-P2, G3-P1)	Replace task completion with narrative choice as the primary structural mechanic	Five story modules with branching choice points; no explicit task list or badge system	PLAY Narrative correlated with Flourishing ($r = 0.568$); P44, P45, P51 reported narrative alignment with personal stress experiences
Rejection of leaderboards and competitive mechanics as pressure-inducing (Theme 3; G5-P2, G5-P3)	Remove all comparative metrics; no points, streaks, or visible rankings	No leaderboard, streak calendar, or social comparison feature in deployed system	Non-punitive engagement endorsed by P47, P55, P56; P56 described the on-demand model as “very helpful”; HEMA Hedonic correlated with Flourishing ($r = 0.601$)

Phase II Theme	Design Decision	System Element	Phase III Evidence
Wellness tools must not become obligations; notifications and mandatory check-ins cause anxiety (Theme 6; G1-P1, G1-P2, G1-P3)	No mandatory daily check-ins; reflective activities framed as optional; brief frictionless onboarding	Optional journaling and mini-game; no push notification requirement; tutorial explicitly states no need to remember everything	Day 5 EMA compliance rose to 78%; Worth It increased from 3.75 to 4.21 across the week, the opposite of declining compliance seen in obligation-oriented systems
Mood tracking over-represents distress; reflection should go beyond emotional labeling (Theme 5; G1-P2, G2-P2)	Replace daily mood logs with narrative-embedded optional journaling using three modalities	Inscriptions system: text, drawing, and voice prompts embedded at story branch points; no daily mood log	P51 described journaling as providing “an outlet for stress and thinking”; multimodal design endorsed by all six expert evaluators prior to deployment
Collaborative repair and peer-anchored support preferred over parental or evaluative companions (Theme 2; G3-P1, G3-P2, G5-P1)	Design NPCs as non-directive guides; peer voice distinct from clinical authority	The Curator: non-evaluative narrative scaffold; Poppy: peer-anchored support figure speaking from personal experience	P44 and P51 described Curator reveal as therapeutically significant; P46 identified with Poppy’s described experience; Curator legibility gap noted by P47 and P56

Phase II Theme	Design Decision	System Element	Phase III Evidence
Preference for flexible systems adapting to shifting emotional states; mental health is not one-size-fits-all (Theme 3; G2-P2)	Implement Dynamic Difficulty Adjustment in Mind Defender; auto-advancing pacing with speed selection	DDA system monitors behavioral indicators and adjusts enemy speed and power-up frequency; three pacing options (slow, medium, fast)	P43 reported in-story choices tracking real-time stress state; P47 noted relaxation effect distinguishable from other games; GAMEX Agency correlated with ETES Authenticity ($r = 0.842$)
Avatar and character enactment lower social risk and enable rehearsal of coping approaches (Theme 1; G1-P1, G2-P1, G6-P3)	Ground player identity in avatar reply bubbles at choice points; maintain narrative-world framing throughout	Player avatar appears beside reply bubble after each choice; no external research rationale surfaced to player	P44 described in-game expression as more authentic than real-world constraints would allow; P51 described avatar distance as enabling self-reflection; Eudaimonic HEMA-r predicted ETES Overall ($r = 0.718$)

Across seven traced decisions, five showed direct Phase III validation through either significant correlations or participant qualitative endorsement. The Curator legibility gap represents the clearest case of partial validation: the character’s supportive function was recognized by some participants but not consistently legible to others, identifying a specific design target for future iteration. The Inscriptions system received qualitative endorsement from participants and all expert evaluators but low interaction rates in the deployment data, suggesting that embedding alone is insufficient to guarantee engagement with optional reflective activities.

5.11 Summary of Findings

Table 5.17 presents all statistically significant findings across the pre/post, EMA, and correlation analyses for Phase III.

Table 5.17: Summary of All Significant Phase III Findings

X	Y	<i>r</i> or Δ	Source
<i>Pre/Post Change Scores (N = 15)</i>			
Pre PSS	Post PSS	$\Delta = -2.47$	Pre/Post
Pre Flourishing	Post Flourishing	$\Delta = +1.93$	Pre/Post
Pre HEMA-r	Post HEMA-r	$\Delta = +0.36$	Pre/Post
<i>EMA Daily Trends (N = 47)</i>			
Day 1 Stress	Day 5 Stress	$\Delta = -1.82$	EMA
Day 1 Coping	Day 5 Coping	$\Delta = +1.50$	EMA
Day 1 Worth It	Day 5 Worth It	$\Delta = +0.46$	EMA
<i>EMA Internal (N = 47)</i>			
Worth It	Coping Confidence	0.767	EMA
<i>EMA-Survey Integrated (N = 15)</i>			
Avg EMA Stress	Post PSS	0.734	Integrated
Avg EMA Worth It	ETES Overall	0.847	Integrated
Avg EMA Worth It	GAMEX Agency	0.517	Integrated
Avg EMA Coping	ETES Overall	0.691	Integrated
<i>Post-Survey Correlations (N = 15)</i>			
PLAY Overall	Flourishing	0.507	Post-Survey
HEMA Hedonic	Flourishing	0.601	Post-Survey
GAMEX Agency	ETES Authenticity	0.842	Post-Survey
PLAY Emotion	ETES Overall	0.781	Post-Survey
GAMEX Agency	PLAY Emotion	0.754	Post-Survey
PLAY Narrative	Flourishing	0.568	Post-Survey
Eudaimonic	ETES Self-Act.	0.727	Post-Survey
Eudaimonic	ETES Overall	0.718	Post-Survey
Eudaimonic	GAMEX Agency	0.633	Post-Survey
HEMA Overall	GAMEX Overall	0.520	Post-Survey

Across all three Phase III data sources, the findings are broadly consistent in direction. Perceived stress decreased and flourishing increased for the majority of matched participants. In-the-moment stress declined across the week while coping confidence rose, and participants found the experience increasingly worthwhile over time. The strongest predictors of post-survey wellbeing and transformative experience were agency,

emotional resonance with the narrative, and the moment-to-moment sense that engagement was worth the participant's time — a convergent pattern across the EMA and correlation data that points toward a plausible experiential mechanism even in the absence of a control condition.

5.12 Limitations

The matched sample of $N = 15$ is small for Pearson correlation analysis; moderate effects may not have reached significance and findings should be treated as preliminary pending replication. The study lacked a control group or comparison condition; pre/post changes cannot be causally attributed to engagement with *The Archive* and may reflect natural stress variation, regression to the mean, or time-period effects independent of the intervention. EMA compliance was moderate (52.2%), and daily response counts on Days 2 and 3 were low enough that day-level estimates are unstable; day-by-day comparisons should be interpreted cautiously. Three participants with elevated pre-survey PSS scores (29 and 36) did not complete the post-survey; their outcomes could shift group statistics in either direction when collected. All findings rely on self-report data, which is subject to social desirability bias, demand characteristics, and retrospective distortion. Finally, post-survey data were collected at a single time point; whether any observed changes in stress or flourishing persisted beyond the study period is unknown. A follow-up assessment at two to four weeks post-study would be necessary to evaluate sustained effects.

Chapter 6

Discussion

This chapter interprets the findings from Phase III in relation to the research questions that guided this work, the prior literature reviewed in Chapter 2, and the design and methodological decisions documented in Chapters 3 and 4. The discussion is organized around four themes that cut across the three research questions: the preliminary evidence for stress reduction and wellbeing support, the role of agency and narrative quality as design mechanisms, the alignment between participatory design values and observed outcomes, and the contributions and limitations of a Research through Design approach to this problem space. The chapter closes with implications for future design and research.

6.1 Stress, Flourishing, and the Case for Modest Claims

The pre/post data from Phase III showed a directional decrease in perceived stress and a more consistent increase in flourishing for the majority of matched participants. Average PSS scores declined by 2.47 points, with 60% of participants showing improvement. Flourishing scores increased by an average of 1.93 points, with 73% of participants improving and only one showing any decline. These findings warrant careful interpretation. The study had no control group, a sample of fifteen matched participants, and a five-day deployment window. None of these characteristics support strong causal claims, and they are not offered here as such.

What the findings do support is a preliminary characterization of The Archive as a

flourishing-supportive experience for the majority of participants who engaged with it over a short period of time. The directional consistency of the Flourishing results — 73% improved, 7% worsened, with a standard deviation of change scores of only 2.05 — is notable for a study of this size and an intervention of this brief duration. Flourishing is typically understood as a multi-domain, relatively stable indicator of psychological wellbeing that assesses meaning, engagement, positive relationships, and self-esteem (**diener'new'2010**). Short-term interventions do not typically move it reliably. The fact that movement in the positive direction was observed in nearly three-quarters of participants suggests that the experience was touching something real about participants' psychological orientation, even within a five-day window.

This interpretation is consistent with prior research on eudaimonic game experiences and meaningful interaction, which suggests that the value of narrative play for wellbeing does not necessarily require extended engagement or measurable skill development (**mekler2016; rusch2017**). Rather, it may emerge through brief but genuine encounters with meaningful content — moments where a player feels that what they are doing matters and reflects something true about their experience. The Phase II focus group data documented in Chapter 3 showed that participants consistently described wanting experiences that acknowledged fluctuation, allowed pause, and felt personally relevant rather than prescribed. The EMA data from Phase III suggests that this quality was present: Worth It scores increased across the week rather than declining, and coping confidence rose even as the narrative content became more emotionally demanding. These patterns are not consistent with the burnout and disengagement that prior research has associated with compliance-oriented wellness systems (**hoffmann'gamification'2017**).

The PSS findings, while directionally positive, are more mixed and less consistent than the Flourishing results. Five participants showed modest increases in perceived stress over the study period, and the standard deviation of PSS change scores (4.69) was more than twice that of Flourishing change scores (2.05). This variability may reflect several things. Stress is more acutely sensitive to immediate life circumstances than flourishing, and a five-day window captures whatever week participants happened to be having. Two participants who began the study with the highest pre-survey PSS scores among the matched sample showed the largest decreases, which is consistent with

regression to the mean and makes it difficult to attribute those changes to the intervention. Conversely, the standout cases of improvement — participants who decreased by 8, 10, and 12 PSS points respectively — also tended to have elevated starting scores, so the same interpretive caution applies.

What the PSS data can support is the claim that engagement with The Archive was not associated with stress increases in any systematic way. Participants who showed PSS increases showed them in magnitudes of one to three points, which falls within typical week-to-week fluctuation. None showed increases that would suggest the content was aversive or distressing in a way that carried into the post-survey. This is meaningful given that the narrative content of the game is explicitly stress-themed and includes emotionally confrontational material, particularly in the Poppy and Contract Fulfilled modules. The absence of systematic harm is itself an interpretively relevant outcome for a wellness system designed to engage with difficult content rather than avoid it.

6.2 Agency as the Central Design Mechanism

The most theoretically significant finding in Phase III is the cluster of relationships centered on agency. The strongest correlation in the entire post-survey dataset was between GAME-X Agency and ETES Authenticity ($r = 0.842$, $p < 0.05$). GAME-X Agency also significantly predicted PLAY Heuristics Emotion ($r = 0.754$), and PLAY Heuristics Emotion in turn predicted ETES Overall ($r = 0.781$). Taken together, these relationships suggest a pathway from player agency through emotional connection to transformative experience — and that this pathway was operating in Phase III data in a strong and consistent way.

This finding connects directly to what Phase II participants described as the most important quality of an alternative to existing wellness applications. Across eleven focus groups, participants returned repeatedly to the idea of meaningful choice as the distinguishing feature between experiences that felt supportive and those that felt coercive. As one participant put it, “I always choose your own adventures when it has a real impact on what happens” (**imx**). Choice was not described as cosmetic or decorative. It was described as authorship — the sense that what you decided would genuinely shape what happened and what the story meant. The Phase I survey data corroborated this:

narrative-driven and autonomy-supporting mechanics were rated most favorably, while competitive and evaluative mechanics received the lowest desirability ratings (**acmih**).

The GAME-X Agency findings from Phase III suggest that this design principle was successfully instantiated in the deployed system. Participants who felt in control of the narrative also felt more emotionally connected to it and reported greater transformation through it. The tight correlation between agency and authenticity is particularly meaningful in the context of a wellness-oriented system. Authenticity — the sense that an experience allowed genuine self-expression and exploration of real values and feelings — is not a quality that can be prescribed or produced through directive instruction. It emerges when participants feel free to determine what the experience means to them, rather than being told what it should mean. The design of The Archive, which embedded choices at every level of the narrative and framed those choices as different ways of engaging rather than correct or incorrect responses, appears to have supported this quality of experience for the majority of participants.

This interpretation is consistent with research on eudaimonic game design, which argues that meaningful engagement emerges not from performance or achievement but from agency, interpretation, and the sense that what one does in a game has genuine consequences within a world that feels coherent and responsive (**mekler2016a**). It is also consistent with prior work on reflective informatics, which emphasizes interpretation over instruction as the central design goal for systems intended to support sensemaking (**baumer2014**). The Archive’s non-evaluative framing — no scores, no failure states, no streak mechanics, no performance metrics — appears to have created the conditions in which agency could function as a pathway to authentic experience rather than being crowded out by evaluative pressure.

The connection between agency and emotional resonance ($r = 0.754$) is also worth unpacking. The PLAY Heuristics Emotion subscale captures the degree to which an experience elicited genuine emotional responses and felt emotionally relevant to the player’s own life. The strong correlation between feeling in control and feeling emotionally moved suggests that these qualities reinforce each other in narrative play. When players feel that their choices genuinely shape the story, they invest emotionally in those choices and their consequences. When they feel emotionally invested, the narrative becomes personally relevant rather than abstractly thematic. This dynamic is consistent

with narrative theory that frames meaningful choice as a mechanism of emotional identification and co-authorship (**murray1997; isbister2016**).

The scoping review conducted as part of Phase II’s background work found that across 66 studies of mental health-oriented games, play-oriented approaches that combined immediate feedback with reflective or narrative elements were more consistently described as meaningful and supportive than those relying exclusively on reward loops (**yearning**). The Phase III data adds empirical weight to this pattern by demonstrating that agency and narrative quality were the specific design qualities most strongly predictive of both emotional transformation and psychological wellbeing in this sample.

6.3 Narrative Quality, Worth, and Wellbeing

Two additional significant findings from Phase III extend the picture of how The Archive’s design qualities connected to outcomes. First, PLAY Heuristics Narrative was significantly correlated with Flourishing ($r = 0.568, p < 0.05$). Participants who rated the quality of the narrative more highly also reported higher psychological wellbeing at post-survey. Second, the integrated analysis showed that average EMA Worth It was the strongest predictor of ETES Overall across the entire dataset ($r = 0.847, p < 0.05$). Participants who consistently felt that engaging with the game was worth their time in the moment were dramatically more likely to report transformative experience at post-survey.

These findings are related but distinct. The PLAY Narrative-Flourishing relationship suggests that the structural and aesthetic qualities of the narrative — its coherence, depth, and the degree to which it presented stress in a way that felt resonant and navigable rather than overwhelming or trivial — contributed to participants’ psychological wellbeing. This is consistent with a body of work showing that narrative experiences can support emotional processing and sensemaking in ways that contribute to wellbeing beyond their entertainment value (**pennabaker1997; rusch2017**). The Phase II findings documented that participants specifically imagined stress as inhabitable environments — haunted libraries, looping corridors, vanishing spaces — because spatial metaphor gave stress a form that could be moved through rather than optimized away (**imx**). The narrative structure of The Archive was built directly from these metaphors.

The correlation between narrative quality and flourishing suggests that this translation was felt by participants as meaningful rather than arbitrary.

The EMA Worth It finding operates at a different level. Rather than capturing how participants retrospectively evaluated the narrative quality of the experience, it captures whether they felt the engagement was worthwhile in the immediate moment of completing each module. The fact that this in-the-moment sense of worth was the strongest predictor of transformative experience across the entire study suggests that the pathway from narrative engagement to transformation runs primarily through subjective value in the present moment rather than through any particular structural feature of the design. When participants found value in what they were doing as they were doing it, transformation followed. When they did not, it did not, regardless of other design qualities.

This finding has implications for how wellness games and narrative systems should be evaluated. Retrospective measures of design quality, game experience, and transformative outcome capture important dimensions of the experience, but they may miss the moment-to-moment process through which these outcomes are actually produced. The EMA data suggests that participants who found the experience consistently worthwhile across multiple module completions — not just one memorable one — were significantly more likely to report transformation. This speaks to the value of sustained, repeated engagement with a narrative experience rather than a single encounter, and it supports the design decision to structure *The Archive* as a five-module sequential narrative rather than a standalone interactive piece.

The strong correlation between EMA Worth It and EMA Coping Confidence ($r = 0.767$ at the response level, across 47 responses) adds a further dimension. In the moments when participants found the engagement worthwhile, they also felt more capable of coping with stress. In-the-moment stress showed only weak, non-significant correlations with both measures. This pattern suggests that what made engagement feel worth participants' time was not primarily that it reduced their stress in the moment — it was something about the quality of the engagement itself that simultaneously supported their sense of coping capability. This is consistent with a self-efficacy framing in which the experience of working through a difficult narrative, making choices with genuine

consequences, and arriving at some form of meaning or resolution builds the participant's sense that they are capable of navigating their own stress (**bandura1997**). The narrative form may function as a low-stakes rehearsal space for coping — a place where the player can practice moving through difficulty without the costs and consequences of real-world stressors.

6.4 Participatory Design Values and Motivational Alignment

The correlation analysis relevant to RQ3 showed that eudaimonic motivation was a consistent predictor of multiple positive outcomes in Phase III. Participants who engaged with the study seeking growth, meaning, and personal development were significantly more likely to report self-actualization, overall transformative experience, and a stronger sense of agency during play. This pattern has two possible interpretations that are not mutually exclusive.

The first interpretation is that the participatory design process successfully aligned the system's design values with the motivational orientations of participants who benefit most from experiences of this kind. The Phase II focus groups and the Phase I survey both pointed consistently toward the same design priorities: narrative over task completion, optional reflection, no evaluative framing, player agency as the central mechanism. These are all qualities that should be especially meaningful to participants who are intrinsically motivated toward growth and reflection. If the participatory process produced a system that speaks the language of eudaimonic motivation, then participants with that motivational profile should be disproportionately likely to find it meaningful — and that is what the data shows.

The second interpretation is that participants with eudaimonic orientations are generally more likely to find any reflective experience transformative, independent of how well it was designed. Dispositional openness to growth and meaning might predict ETES scores across a wide range of interventions, and the relationship observed in Phase III might not be specific to this system. Distinguishing between these interpretations would require a comparative design in which the same participants engage with systems varying in how well they align with eudaimonic motivational values.

Both interpretations, however, point toward the same practical implication: the participatory design approach did not produce a system that was indifferent to the motivational orientations of its users. Whether through intentional alignment or through the general fitness of eudaimonic experiences for eudaimonic users, the system appears to have been particularly meaningful for participants who came to it seeking something beyond entertainment. This is consistent with the design philosophy that emerged from Phase II, which positioned The Archive as a space for exploration and interpretation rather than an application for managing or optimizing stress. Participants who wanted to be managed or optimized may have found the experience less resonant. Participants who wanted to understand and navigate their stress found it meaningful.

The HEMA-r Hedonic motivation finding complicates this picture in a productive way. Hedonic motivation — seeking enjoyment, relaxation, and pleasant experience — was significantly correlated with Flourishing ($r = 0.601$) but not with the deeper transformative outcomes measured by the ETES. This suggests that participants who engaged with the system primarily seeking enjoyment did experience a wellbeing benefit — they reported higher flourishing — but were less likely to experience the kind of growth and authentic self-expression that the ETES captures. This distinction is consistent with prior theoretical work differentiating hedonic from eudaimonic wellbeing and experience quality (**waterman1993**), and it suggests that The Archive may be operating differently for participants with different motivational profiles. For hedonic users, it functions as a pleasant, wellbeing-supportive narrative experience. For eudaimonic users, it functions as something closer to a transformative encounter with their own stress experience.

This motivational differentiation may itself be a feature of the design rather than a limitation. The Phase II participants did not describe a single type of engagement as desirable; they described a range of ways that stress could be navigated. Some wanted depth and confrontation. Others wanted lightness and relief. The Archive’s narrative structure accommodates both orientations: players can engage with the emotional content as deeply or as lightly as they choose, skip reflective prompts, replay modules, or engage with the mini-game without completing the journal. The quantitative data suggests that participants at both ends of the motivational spectrum experienced some benefit, with the deeper benefits accruing to those who were already oriented toward

growth.

6.5 The EMA as a Research and Design Tool

A secondary contribution of Phase III is the validation of the embedded EMA as a research instrument and as a potential design feature. The convergent validity evidence — a correlation of 0.734 between average EMA stress and post-survey PSS — is strong for a three-item, in-the-moment measure embedded within a game application. This suggests that the EMA was successfully capturing the same underlying stress construct as the validated PSS-10, and that brief embedded assessments of this kind can function as real-time proxies for longer validated instruments in naturalistic game deployment studies.

The EMA data also surfaced patterns that would have been invisible to pre/post survey analysis alone. The Day 3 dip in coping confidence, the U48 stress spike trajectory, the increasing compliance rate across the week — none of these would have been detectable from a simple pre/post design. These patterns suggest that the experience of The Archive across the five-day period was not uniform or linear. It involved moments of elevated demand, particularly around the Poppy module’s disclosure mechanics, followed by recovery and stabilization. This temporal texture is consistent with the theoretical framing of the game as a space for moving through stress rather than eliminating it, but it could only be documented through real-time measurement.

The strong EMA Worth It-Coping relationship ($r = 0.767$) and the EMA Worth It-ETES relationship ($r = 0.847$) also have practical implications for the design of future studies. Worth It — a single-item measure of perceived engagement value at the moment of completion — appears to be an unusually powerful predictor of transformative experience outcomes. If this finding replicates in future studies, it would suggest that brief, embedded worth-it assessments could serve as efficient proxies for more burdensome post-study measures of transformative experience quality. Researchers studying narrative wellness systems could use real-time worth-it ratings to identify which modules, story paths, or interaction patterns are most likely to support the deeper outcomes that motivate the design.

6.6 Research through Design: Contributions and Limitations

This project operated within a Research through Design framework ([zimmerman2007](#)), which positions designed artifacts as forms of inquiry rather than as interventions to be validated against predefined outcomes. The Archive is presented here as a research artifact — a designed object that embodies hypotheses about how narrative play, participatory design, and reflective technology might intersect in the domain of college student stress — and the findings from Phase III are interpreted as evidence bearing on those hypotheses rather than as proof of effectiveness.

This framing is important for understanding what the study does and does not claim. It does not claim that The Archive is a clinically effective intervention for stress management. The evidence base — fifteen matched participants, no control group, five days of engagement — is insufficient for claims of that kind, and no such claims are made. What the study claims is that the design decisions made in response to participatory findings produced a system that, in preliminary evaluation with a small sample, showed directionally consistent patterns of stress reduction and flourishing improvement, that agency and narrative quality emerged as the dominant experiential mechanisms, and that the in-the-moment sense of worth during engagement was the single most powerful predictor of transformative experience. These are claims about how the designed system functioned and what it revealed about the design space, not claims about clinical effectiveness.

The RtD framing also clarifies the relationship between the five stages of this project. Phases I and II generated the design knowledge that informed Phases III and IV (the application development and expert evaluation). Phase V (the usability study documented here) generated evidence about how the system functioned in practice, which in turn informs future design iterations. Each phase was not simply a step toward a finished product but an opportunity to learn something about the problem space — about how students articulate stress through narrative, about what design qualities they associate with support and pressure, about how agency and emotional resonance operate in narrative play, about what the EMA can and cannot capture in a naturalistic deployment. The archive of knowledge accumulated across these five stages constitutes

the primary contribution of the research, alongside the designed artifact itself.

Within this framework, the expert evaluation documented in Chapter 3 served an important function. The ten expert evaluators who engaged with the system through think-aloud protocols and structured interviews provided domain-specific insight into the narrative design, clinical appropriateness, and usability that informed refinements before Phase V deployment. The themes emerging from that evaluation — around the pacing of emotional disclosure, the clarity of the Curator’s onboarding role, and the optional framing of the reflective activities — shaped the final version of the system that participants engaged with in Phase V. The RtD process is not linear, and the expert evaluation represents a point at which design knowledge flowed back into the artifact before it was evaluated with participants.

6.7 Limitations and Future Directions

Several limitations of this work should shape how the findings are interpreted and how future research is designed. The sample size remains the most significant constraint. With fifteen matched participants, the study is adequately powered to detect large effects but not medium ones, and several relationships in the data that trended in theoretically meaningful directions did not reach significance at this threshold. A larger sample of 30 or more participants, recruited across a longer enrollment window, would provide substantially more confidence in the patterns observed here.

The absence of a control group prevents causal attribution of the observed pre/post changes. Future research should incorporate a waitlist control or active comparison condition — for example, comparing The Archive to a commercially available wellness application such as Headspace or Calm — to assess whether the narrative approach produces outcomes different from those achievable through existing tools. This comparison would also speak directly to the Phase II finding that participants described existing applications as counterproductive and evaluative, and it would test whether the design principles embedded in The Archive produce meaningfully different experiential outcomes.

The EMA compliance rate of 52.2%, while typical for naturalistic EMA research,

leaves room for improvement. Future iterations of the study should consider push notifications within the application following each module completion, brief in-app reminders calibrated to participants' typical engagement times, or small additional incentives for maintaining daily compliance. The low compliance on Day 2 in particular suggests that early-week engagement may be a vulnerability in the current design, and that onboarding support could be strengthened to help participants establish their engagement pattern before the more emotionally demanding content begins in later modules.

The study's population was limited to undergraduate and graduate students at a single mid-sized public university in the United States. Stress is experienced differently across populations, cultural contexts, and institutional settings, and findings generated with a relatively homogeneous college student sample may not generalize to other groups or other forms of stress. Future research should examine how narrative approaches to stress support function in other populations — including community college students, graduate students in particularly high-pressure programs, or adult learners whose stressors differ from traditional undergraduates.

The qualitative dimensions of the Phase III experience remain underdeveloped in the current analysis. Journal entries, if fully analyzed, would provide insight into what participants reflected on, how they framed their stress through the Archive's narrative language, and whether the reflective activities connected to identifiable patterns of meaning-making. Voice memo entries in particular — which participants recorded in real time within the narrative context — would provide a form of data qualitatively different from written reflection. Future analysis of this data, coded for themes consistent with the theoretical framework, would substantially enrich the quantitative picture presented here.

Finally, the study measured outcomes at a single post-survey timepoint immediately following the five-day deployment. Nothing is known about whether the observed changes in stress and flourishing were sustained in the weeks following the study, or whether participants continued to engage with the application, applied the coping strategies modeled in the narrative, or returned to the reflective prompts they may have skipped during the study period. A follow-up assessment at two to four weeks post-study would begin to answer these questions and would be an important next step in understanding the longitudinal impact of narrative-based wellness systems.

6.8 Summary

This chapter has interpreted the Phase III findings in relation to the study’s research questions and the broader literature on narrative games, participatory design, and digital mental health. The preliminary quantitative evidence suggests that The Archive produced directionally positive outcomes for the majority of participants who engaged with it, with flourishing improvement showing particularly strong consistency. Agency emerged as the dominant design mechanism, predicting emotional resonance and authentic self-expression more strongly than any other design quality. The in-the-moment sense that engagement was worth participants’ time was the most powerful predictor of transformative experience across the entire dataset, suggesting that the pathway from narrative engagement to transformation runs through subjective moment-to-moment value rather than through any single retrospectively evaluated design feature.

These findings are consistent with the design philosophy developed through the participatory process in Phases I and II, which positioned The Archive as a space for exploration and interpretation rather than for behavioral optimization or compliance. They are also consistent with prior literature on eudaimonic game design, reflective systems, and the limitations of compliance-oriented wellness applications. The study makes a preliminary case that narrative play, designed through participatory methods and grounded in how students themselves understand and narrate their stress experiences, can support meaningful wellbeing outcomes even in a brief deployment — not by telling participants what to feel or do, but by creating the conditions in which they can figure it out for themselves.

Chapter 7

Conclusion

7.1 Overview

This thesis investigated how participatory design methods can inform the development of a narrative-based serious game for college student stress, and what such a system can reveal about the relationship between interactive narrative, player agency, and psychological wellbeing. The work unfolded across five stages: an experiential survey of student gamification preferences, a series of narrative focus groups using a collaborative roleplay probe, the design and development of *The Archive*, an expert evaluation, and a five-day deployment study with college students. Each stage generated knowledge that shaped the next, and the accumulated findings across all five stages constitute the primary contribution of this work.

This chapter synthesizes those contributions, situates them within the broader conversation about digital wellness technology and serious game design, and identifies the directions in which this line of inquiry should continue. It does not restate the findings in detail — those are presented in Chapter 5 and interpreted in Chapter 6 — but draws from them to make the larger claims this research supports.

7.2 What This Work Set Out to Do

College students experience stress as a persistent, fluctuating, and interpretive condition shaped by academic pressure, identity, belonging, and uncertainty (**misra'college'2000**;

bruner1990). Research consistently documents that students underutilize institutional mental health services and frequently turn instead to commercially available digital wellness applications (**lipson'increased'2019**). Those applications, however, have been widely critiqued — and were critiqued by participants in this study — for framing stress management through compliance-oriented mechanics that replicate the evaluative pressures of academic life rather than relieving them (**hoffmann'gamification'2017; bogost2007**).

This thesis proceeded from the observation that the problem with most wellness applications is not that they fail to engage users — many succeed at that — but that what they engage users in doing is not particularly supportive. Streaks, reminders, points, and progress metrics produce consistency, but consistency is not the same as care. Students who miss a check-in feel guilty. Students who break a streak feel as though they have failed at something that was supposed to help them. The design logic of compliance-oriented systems maps poorly onto the reality of stress as an experience, which is irregular, contextual, and resistant to behavioral scheduling.

The design question this thesis pursued was whether a different approach — one grounded in narrative play, participatory design, and player agency — could produce a different kind of experience. Not one that manages stress by regulating behavior, but one that supports reflection on stress by giving it a form that can be moved through, interpreted, and made sense of. *The Archive* is this thesis's answer to that question, and the findings from its deployment provide a preliminary empirical basis for evaluating it.

7.3 Contributions

7.3.1 Empirical Understanding of Student Stress Expression

The first contribution of this work is an empirical account of how college students use narrative metaphor and collaborative world-building to articulate and make sense of academic stress. Across eleven focus groups involving thirty participants, students consistently represented stress through spatial and environmental metaphors: haunted libraries, looping corridors, vanishing spaces, and creatures that grew larger when left unattended. These metaphors were not incidental. They were structurally meaningful — stress became something inhabitable, something that could be moved through rather

than measured, and something that was linked to avoidance and incompleteness rather than to external failure.

This finding extends and contextualizes prior research on academic stress by documenting the specific forms that stress takes in students' narrative imagination, and by demonstrating that collaborative storytelling is a productive method for surfacing these forms. Students who might have struggled to articulate their stress through direct self-report were able to express it through fiction without losing the personal resonance that made their accounts analytically meaningful (**sanders'co-creation'2008; zimmerman2007**). The focus group data also revealed that students do not experience stress as a problem to be solved. They experience it as something to be navigated — a condition with texture, variability, and multiple possible responses, none of which is definitively correct. This framing directly informed the design philosophy of *The Archive*, which treats stress as something to move through rather than something to optimize away.

7.3.2 A Systematic Critique of Wellness Application Design

The second contribution is a systematic critique of common wellness application design patterns from the perspective of students' lived experience. Participants in Phase II did not simply describe what they wanted. They identified with precision what existing applications were doing wrong and why. Streaks introduced guilt. Points felt disconnected from personal meaning. Daily check-in structures added obligations to already overloaded schedules. Aesthetic customization without narrative grounding felt hollow. The findings from Phase I reinforced this critique at the level of gamification mechanics: narrative-driven and autonomy-supporting features were rated most favorably, while competitive and performance-tracking mechanics were rated lowest among participants who identified as experiencing stress.

This critique is not new in the literature on digital mental health (**hoffmann'gamification'2017; bogost2007**), but it is substantiated here with direct participant accounts across both quantitative preference ratings and qualitative narrative data. The synthesis across Phases I and II provides a more granular empirical basis for the design claim that compliance-oriented mechanics are not just neutral or ineffective in wellness contexts — they are actively misaligned with how students experience and prefer to engage with

stress. This has implications for the broader field of digital mental health design, which continues to rely heavily on behavioral reinforcement frameworks that were developed for habit formation and productivity rather than for care.

7.3.3 Design Principles for Narrative Wellness Systems

The third and most directly practical contribution is the set of design principles that emerged from the participatory process and were tested through the deployment of *The Archive*. These principles are not prescriptive rules. They are design values — orientations that shaped decisions about what to include, what to exclude, how to frame interaction, and how to relate to the player. They are offered here as implications for other designers working at the intersection of narrative, play, and wellbeing.

Narrative over task completion. Progress in *The Archive* is represented through story movement and symbolic items rather than through scores, levels, or streaks. Players know where they are in the narrative because the story tells them, not because a dashboard displays their performance. This framing positions the player as a reader and participant in an unfolding world rather than as a performer being evaluated. The Phase III data suggests this framing supported emotional connection and authentic self-expression, as indexed by the significant relationship between PLAY Heuristics Narrative and Flourishing and between GAME-X Agency and ETES Authenticity.

Agency as mechanism, not decoration. The choices embedded in *The Archive* are not cosmetic variations on a predetermined path. They shape the format of reflective activities, the tone of subsequent passages, the items players carry, and the way certain narrative moments unfold. Players who chose to notice the smell of the library entered a different reflective modality than players who chose to notice the shelves. Players who disclosed different stress themes to Poppy received different coping advice and different mini-game items. This is agency in the service of personal relevance rather than agency as the simulation of control. The Phase III data showed that this quality of agency — the sense that choices matter and that the experience responds genuinely to them — was the strongest predictor of authentic self-expression and emotional connection with the narrative.

Optional reflection without evaluation. Journaling, drawing, and voice memo

prompts in *The Archive* are embedded within the story rather than presented as separate tasks. They are optional. They are not graded or evaluated. Players who skip them can access them later through the Resource Center. Players who complete them receive no reward or acknowledgment beyond the narrative moment that prompted them. This framing was a direct response to focus group participants who described reflection as something that should emerge from narrative engagement rather than be imposed as a requirement. The EMA data suggests it worked: participants who found the engagement worthwhile in the moment were dramatically more likely to report transformative experience at post-survey, and Worth It was not correlated with whether participants completed optional reflective activities on any given day.

Non-directive guidance. The Curator and Poppy, the two supporting characters in *The Archive*, were explicitly designed to avoid the instructive, corrective, and evaluative posture that focus group participants associated with wellness applications and with institutional mental health resources. The Curator contextualizes without prescribing. Poppy advises from personal experience rather than professional authority. Neither character tells the player what to feel, what their choices mean, or what they should do differently. This design choice was grounded in Phase II data showing that students wanted to feel accompanied rather than managed, and in research on reflective systems arguing that interpretation rather than instruction should be the central interactional goal for systems intended to support sensemaking (baumer2014).

7.3.4 The Archive as a Research Artifact

The fourth contribution is *The Archive* itself, understood as a research-through-design artifact (zimmerman2007). The system embodies and tests a set of hypotheses about how narrative play, participatory design, and reflective technology can intersect in the domain of college student stress. It is not offered as a clinical intervention or as a finished product. It is offered as a designed object that makes those hypotheses tangible and testable, and that generates knowledge about the problem space through its deployment and evaluation.

As a research artifact, *The Archive* contributes to the existing conversation in HCI and serious game design about how to build wellness systems that support reflection rather than regulation. It extends prior work on eudaimonic game design (mekler2016;

rusch2017), narrative scaffolding for emotional reflection (stratton2025stalls), and participatory design for sensitive experiential domains (sanders'co-creation'2008; bodker2004) by instantiating these theoretical commitments in a deployed mobile system and evaluating them with participants. The preliminary findings from Phase III provide an empirical basis for continued development and more rigorous evaluation of this design approach.

7.3.5 Methodological Contributions

The fifth contribution is methodological. This thesis demonstrated that a customized narrative roleplay probe — adapted from a tabletop game format — can function as an effective participatory design instrument for surfacing how students understand and narrate stress. The Blackthorn University fictional setting, the character and relationship card structure, and the Tilt mechanic that introduced disruptive events all served to distance participants from direct self-disclosure while keeping the content personally resonant. The data generated through this method was richer in narrative texture and more surprising in its specific metaphorical content than direct questionnaire-based research would have been able to produce (gaver'design'1999; sanders'co-creation'2008).

The embedded EMA system also represents a methodological contribution to the study of narrative wellness systems. The strong convergent validity between average EMA stress and post-survey PSS ($r = 0.734$) suggests that brief, in-application real-time measures can function as valid proxies for longer validated instruments in naturalistic game deployment contexts. The finding that EMA Worth It was the strongest predictor of transformative experience outcomes across the entire dataset ($r = 0.847$) suggests that in-the-moment perceived value of engagement is a particularly efficient signal for researchers studying whether a wellness game is producing the experiential conditions for meaningful outcomes. Future studies of narrative wellness systems should consider embedding brief EMA measures as both a research tool and as a source of data that would be impossible to collect through pre/post designs alone.

7.4 Positioning Within the Broader Field

This work sits at the intersection of three active research areas: digital mental health technology, serious games and interactive narrative, and participatory design in HCI. Each of these areas has contributed conceptual and methodological resources to the project, and the findings contribute back to each of them.

Within digital mental health research, this work reinforces and deepens the critique of compliance-oriented design while adding empirical support for narrative and autonomy-supporting alternatives. Prior reviews of digital mental health applications have documented the prevalence of reinforcement-based mechanics and the gap between what those mechanics produce and what users in stress contexts actually need (**hoffmann'gamification'2017**; **linardon'attrition'2020**). This thesis provides direct student accounts of why that gap exists, and preliminary quantitative evidence that an alternative approach — narrative, non-evaluative, agency-centered — can produce directionally positive wellbeing outcomes. The findings do not resolve the question of clinical effectiveness, but they establish a stronger empirical basis for pursuing this design direction than has previously existed.

Within serious games and interactive narrative research, this work contributes evidence that agency is not merely a design quality associated with enjoyment or engagement, but a mechanism through which narrative systems can support authentic self-expression and emotional transformation. This is consistent with the broader theoretical claim that play, to be meaningful for wellbeing, must create a bounded space in which exploration, ambiguity, and interpretation are possible rather than being crowded out by performance pressure (**huizinga'homo'2014**; **sicart2014**). The scoping review conducted as part of this project found that across sixty-six studies of mental health-oriented games, the distinction between effective and ineffective systems was not the presence of game mechanics but how those mechanics framed agency, interpretation, and emotional meaning-making (**yearning**). The Phase III data adds empirical weight to this conclusion by identifying agency as the single strongest design predictor of the outcomes that narrative wellness systems aim to support.

Within participatory design in HCI, this work demonstrates the productivity of

narrative-based co-design methods for sensitive health and wellbeing domains. Participants in Phase II were not simply asked what features they wanted. They were invited to create worlds, make choices, and reflect on what those choices meant. The design materials that emerged from that process — spatial metaphors, character-coping alignments, narrative arcs derived from participant-generated stress scenarios — were richer and more specific than what direct elicitation methods would have produced, and they shaped the narrative content of The Archive in ways that were felt by participants in Phase III as meaningful and personally relevant. This confirms prior arguments for participatory and co-design approaches in health technology research (**bodker2004; sanders'co-creation'2008**) and extends them to the specific domain of narrative wellness games.

7.5 Limitations and What Remains to Be Done

This work is an early-stage investigation rather than a mature program of research, and several important questions remain open. The sample size in Phase III limits the confidence of the quantitative findings. The absence of a control group prevents causal attribution of the pre/post changes. The single post-measurement timepoint leaves questions about sustained effects unanswered. And the study's geographic and demographic scope — students at a single mid-sized public university — limits the generalizability of the findings to other populations and institutional contexts.

Future research should address these limitations through larger studies with randomized designs, longer follow-up periods, and more diverse populations. The most important next step is a controlled study comparing The Archive to a waitlist control and, ideally, to a commercially available wellness application, with sufficient sample size to detect medium effect sizes and a follow-up measurement at two to four weeks post-study. This would allow causal claims to be tested and would generate comparative data about whether the narrative approach produces outcomes distinguishably different from what is achievable through existing tools.

Future design iterations of The Archive should also explore the remaining questions raised by Phase III. The Day 3 EMA pattern — a dip in coping confidence associated with the Poppy module's disclosure mechanics — suggests that the emotional arc of

the narrative may benefit from additional scaffolding at the moment of stress disclosure to support participants through the more demanding content before the resolution of later modules. Qualitative analysis of the journal and voice memo entries would provide insight into what participants were actually reflecting on and how they were narrating their stress through the Archive’s framework, which would in turn inform both narrative revisions and the theoretical understanding of what reflective journaling within a game context actually does.

The comparative question about who benefits most from this kind of experience also deserves sustained attention. The Phase III data suggests that participants with eudaimonic motivational profiles found the experience most transformative, while participants with hedonic motivations still experienced wellbeing benefits though of a different character. Understanding the design features that support each motivational profile, and whether the system can be designed to shift motivational orientation over time through the experience, would require both larger samples and more fine-grained measurement of motivational dynamics within and across sessions.

7.6 A Final Note on What It Means to Design for Stress

There is a question embedded in this project that deserves to be made explicit. When we design a wellness application for college students, what are we trying to do? The dominant assumption in the field has been that we are trying to help students manage stress more effectively — to reduce it, regulate it, build habits that prevent it from accumulating, or provide skills for handling it when it arrives. This assumption produces compliance-oriented systems because compliance is the mechanism through which behavioral change is typically achieved.

This thesis has proceeded from a different assumption: that what students need is not primarily to manage their stress differently, but to understand it better. Stress is not only a physiological or behavioral condition. It is an interpretive one. Students make sense of academic pressure through narratives about who they are, whether they belong, whether what they are doing matters, and whether they are capable of the things they are being asked to do. A system that addresses only the behavioral surface of stress without touching the interpretive layer underneath will produce limited outcomes,

because the interpretive layer is where the most consequential work of coping actually happens.

The Archive was designed to reach that layer. Its narrative structure gives stress a form that can be inhabited and moved through. Its choices give players genuine authorship over how that movement unfolds. Its characters model coping as a continuous, imperfect, and personal process rather than a set of skills to be acquired. And its final chapter, in which the creature that has been the player's antagonist reveals itself to be their own shadow-self, makes the interpretive logic of the whole experience explicit: the thing that grows when left unaddressed is not a monster. It is the part of you that knows what needs attending to and cannot get your attention any other way.

This framing positions *The Archive* not as an intervention but as an invitation — to pay attention to what is already there, to give it a form, and to move through it with a little more intention than before. Whether this kind of invitation can be designed at scale, deployed sustainably, and evaluated rigorously enough to inform practice are questions for the next iteration of this work. What this thesis has established is that the invitation itself is worth extending, and that the students who accepted it found it, for the most part, worth their time.

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Appendix A

Glossary and Acronyms

Care has been taken in this thesis to minimize the use of jargon and acronyms, but this cannot always be achieved. This appendix defines jargon terms in a glossary, and contains a table of acronyms and their meaning.

A.1 Glossary

- **Cosmic-Ray Muon (CR μ)** – A muon coming from the abundant energetic particles originating outside of the Earth’s atmosphere.

A.2 Acronyms

Table A.1: Acronyms

Acronym	Meaning
CR μ	Cosmic-Ray Muon

Appendix B

Phase I Survey Instrument

This appendix reproduces the experiential survey used in Phase I. The instrument was administered online via Qualtrics and was organized into five topical sections. All rating items used 5-point Likert-type scales, with the scale anchors reported at the head of each item block. Items marked with *(please specify)* permitted optional open-ended responses. Demographic items were presented at the end of the survey to reduce early dropout and to place eligibility confirmation immediately before submission. The item wording, response options, and scale anchors reproduced here match the version that participants completed.

B.1 Mental Well-Being

1. Do you currently use any self-care apps?

☐ Yes (such as Headspace, Calm, Finch, Forest, Sanvello, Colorfy, or Zombies, Run!)

☐ No *(please specify why not)*: _____

2. Which of the following are your primary sources of stress? *(Select all that apply. Rated 1 = Never, 2 = Sometimes, 3 = About half the time, 4 = Most of the time, 5 = Always.)*

☐ Academic pressure

- ☐ Financial concerns
- ☐ Time management
- ☐ Social life and relationships
- ☐ Physical health
- ☐ Family expectations
- ☐ Loneliness or isolation
- ☐ Other (*please specify*): _____

3. Which activities do you find the most helpful to maintain your mental wellness?
(Select all that apply. Rated 1 = Not at all useful, 2 = Slightly useful, 3 = Moderately useful, 4 = Very useful, 5 = Extremely useful.)

- ☐ Meditation or mindfulness exercises
- ☐ Physical exercise or sports
- ☐ Journaling or self-reflection
- ☐ Talking to friends or family
- ☐ Counseling or therapy
- ☐ Using mental health apps
- ☐ Taking breaks or vacations
- ☐ Setting personal goals
- ☐ Other (*please specify*): _____

B.2 Gamification

1. Rate these game mechanics based on personal preference. *(Rated 1 = One of the worst, 5 = One of the best.)*
- ☐ Point systems and leaderboards (e.g., rankings as in Fortnite, points as in Super Mario Bros.)
 - ☐ Progress bars and levels (e.g., Skyrim progress bar, Enter the Gungeon levels)

- ☐ Personalization (e.g., Baldur's Gate 3 customization, flexibility of play in Minecraft)
- ☐ Story-driven quests (e.g., Baldur's Gate 3 story elements, The Wolf Among Us)
- ☐ Collaborative activities (e.g., Jackbox Games, Among Us, Mario Party)

2. Do these game elements help improve your mood or reduce stress? (*Rated 1 = Strongly disagree, 5 = Strongly agree.*)

- ☐ Relaxing or calming visuals (e.g., Journey, Flower)
- ☐ Engaging stories (e.g., Life is Strange, Red Dead Redemption 2)
- ☐ Creativity (e.g., Minecraft, The Sims)
- ☐ Cooperative play (e.g., It Takes Two, Lethal Company)
- ☐ Collection of achievements, rewards, or items (e.g., Zelda armor and weapons, Animal Crossing furniture, Splatoon 3 clothing)
- ☐ Exploration (e.g., Skyrim, Minecraft, The Legend of Zelda: Breath of the Wild)
- ☐ Physical or movement-based activity (e.g., Beat Saber, Just Dance, Pokémon GO)
- ☐ Problem-solving (e.g., Portal, Tetris, Inside)
- ☐ Humor or lighthearted content (e.g., Untitled Goose Game, Octodad, Goat Simulator)
- ☐ Other (*please specify*): _____

3. How meaningful are the following features in helping you keep your wellness goals in a mental wellness app? (*Rated 1 = Not at all important, 2 = Slightly important, 3 = Moderately important, 4 = Very important, 5 = Extremely important.*)

- ☐ A weekly or monthly mood chart (e.g., Daylio, MyFitnessPal)
- ☐ A calendar marking streaks or consistent days of self-care (e.g., Finch streaks, Duolingo)

- ☐ Progress toward personal wellness goals, such as “You’ve improved focus by 20%” (e.g., Fitbit, Duolingo, Habitica)
- ☐ A journey or quest map that reflects personal achievements (e.g., Zombies, Run!, Pikmin Bloom, The Walk)
- ☐ Other (*please specify*): _____

4. How important do you find features that track and celebrate your progress (e.g., badges for completing daily mood check-ins, progress bars for tracking habits)? (*Rated 1 = Not at all important, 2 = Slightly important, 3 = Moderately important, 4 = Very important, 5 = Extremely important.*)

B.3 Social Support

1. How likely are you to perform self-care challenges in a team setting where you could work on similar goals, such as Facebook groups or HappyFeed grouping pods? (*Rated 1 = Extremely unlikely, 5 = Extremely likely.*)
2. How interested are you in receiving encouragement or motivational messages from others in a group setting? (*Rated 1 = Not interested, 2 = Slightly interested, 3 = Moderately interested, 4 = Very interested, 5 = Extremely interested.*)

B.4 Technology and Engagement

1. Rate your preference on the following game elements in a stress management application. (*Select all that apply. Rated 1 = Do not prefer at all, 2 = Prefer slightly, 3 = Prefer a moderate amount, 4 = Prefer a lot, 5 = Prefer a great deal.*)
 - ☐ Relaxation exercises (e.g., Calm guided meditation, Happier breathing exercises)
 - ☐ Mood tracking (e.g., Daylio, Moodnotes, Moodily)
 - ☐ Personalized recommendations (e.g., Aura changing relaxation on mood)
 - ☐ Community support or chat features (e.g., Talkspace, Sanvello)
 - ☐ Goal-setting and reminders (e.g., Fabulous, HabitBull)

- ☐ Gamified elements such as rewards or challenges (e.g., Habitica, Duolingo)
- ☐ Interactive content such as virtual pets or avatars (e.g., Finch, Amaru)
- ☐ Other (*please specify*): _____

2. Would you find it helpful if the application could adjust content based on real-time information, such as stress level or heart rate, to dynamically change the way you interact with it? Please rate in reference to sensor data from devices such as the Apple Watch, Garmin Watch, or Fitbit. (*Rated 1 = Not at all interesting, 2 = Slightly interesting, 3 = Moderately interesting, 4 = Very interesting, 5 = Extremely interesting.*)

B.5 Demographics and Eligibility

1. Confirm that you are over 18 years of age to participate in this survey.

- ☐ Yes
- ☐ No

2. What gender do you identify with?

- ☐ Male
- ☐ Female
- ☐ Non-binary
- ☐ Prefer not to say

3. What is your current academic year in college?

- ☐ Freshman
- ☐ Sophomore
- ☐ Junior
- ☐ Senior
- ☐ Graduate

4. What is your current major? _____

Appendix C

Phase III Instruments

This appendix reproduces the instruments used during the Phase III field deployment. Four instruments were administered: a pre-study survey completed during the in-person onboarding session, an Ecological Momentary Assessment (EMA) delivered within the application immediately after each completed story module, a post-study survey completed during the in-person Day 6 exit session, and a semi-structured exit interview conducted after the post-study survey. The pre-study and post-study surveys were administered online via Qualtrics. The EMA was delivered within *The Archive* itself. The exit interview was conducted in person and audio-recorded with participant consent. Instruments repeated across pre- and post-study administrations are reproduced in full in the pre-study section and cross-referenced in the post-study section to avoid duplication. The item wording, response anchors, and order reproduced here match the versions as administered to participants.

C.1 Pre-Study Survey

C.1.1 Matching Identifier

Participants created a three-part pseudonymous identifier at the start of the pre-study survey. The same three components were reconstructed in the post-study survey so that pre-study and post-study records could be matched without collecting direct identifiers.

1. As part of this survey, please create a unique identifier so we can match your

pre-survey responses with the post-survey responses. Please enter the following in the box below:

- (a) First letter of mother's first name (A–Z), capitalized: _____
- (b) First letter of your last name (A–Z), capitalized: _____
- (c) Last four digits of your phone number: _____

C.1.2 Gaming Background

1. How often do you play digital games? (*1 = Never, 2 = Less than once a week, 3 = 1–2 times per week, 4 = 3–5 times per week, 5 = Daily.*)
2. How much do you enjoy story-driven games? (*1 = Do not prefer, 2 = Prefer slightly, 3 = Prefer a moderate amount, 4 = Prefer a lot, 5 = Prefer a great deal.*)
3. How often do you read fiction or narrative media? (*1 = Never, 2 = Rarely, 3 = Sometimes, 4 = Often, 5 = Very often.*)

C.1.3 Perceived Stress Scale (PSS-10)

All items ask about the last month. Response anchors: 1 = Never, 2 = Sometimes, 3 = About half the time, 4 = Most of the time, 5 = Always. Items 4, 5, 7, and 8 are reverse-scored.

1. In the last month, how often have you been upset because of something that happened unexpectedly?
2. In the last month, how often have you felt that you were unable to control the important things in your life?
3. In the last month, how often have you felt nervous and stressed?
4. In the last month, how often have you felt confident about your ability to handle your personal problems?
5. In the last month, how often have you felt that things were going your way?

6. In the last month, how often have you found that you could not cope with all the things that you had to do?
7. In the last month, how often have you been able to control irritations in your life?
8. In the last month, how often have you felt that you were on top of things?
9. In the last month, how often have you been angered because of things that happened that were outside of your control?
10. In the last month, how often have you felt difficulties were piling up so high that you could not overcome them?

C.1.4 Flourishing Scale

Response anchors: 1 = Strongly disagree, 2 = Disagree, 3 = Somewhat disagree, 4 = Neither agree nor disagree, 5 = Somewhat agree, 6 = Agree, 7 = Strongly agree.

1. I lead a purposeful and meaningful life.
2. My social relationships are supportive and rewarding.
3. I am engaged and interested in my daily activities.
4. I actively contribute to the happiness and well-being of others.
5. I am competent and capable in the activities that are important to me.
6. I am a good person and live a good life.
7. I am optimistic about my future.
8. People respect me.

C.1.5 Motivation for Destressing Activities (HEMA-R)

Stem: "To what degree do you approach distress activities with each of the following intentions, whether or not you actually achieve your aim?" Response anchors: 1 = None at all, 2 = Very little, 3 = Slightly, 4 = Moderately, 5 = Quite a bit, 6 = Very much, 7 = Extremely.

1. Seeking relaxation?
2. Seeking to develop a skill, learn, or gain insight into something?
3. Seeking to do what you believe in?
4. Seeking pleasure?
5. Seeking to pursue excellence or a personal ideal?
6. Seeking enjoyment?
7. Seeking to take it easy?
8. Seeking to use the best in yourself?
9. Seeking fun?
10. Seeking to contribute to others or the surrounding world?

C.1.6 Demographics

1. Please enter age:

☐ 18

☐ 19

☐ 20

☐ 21

☐ 22

☐ 23

☐ 24

☐ 25

☐ 26+

2. How do you describe yourself?

☐ Male

- ☐ Female
- ☐ Non-binary or third gender
- ☐ Prefer to self-describe: _____
- ☐ Prefer not to say

3. What is your current living situation?

- ☐ On-campus housing
- ☐ Off-campus housing
- ☐ With family
- ☐ Other: _____

4. Have you used counseling services in the past year?

- ☐ Definitely not
- ☐ Probably not
- ☐ Might or might not
- ☐ Probably yes
- ☐ Definitely yes

5. Have you used a mental health or wellness app before?

- ☐ Yes
- ☐ No
- ☐ Why not: _____

C.2 Post-Study Survey

The post-study survey was administered during the in-person exit session on Day 6. It reconstructed the matching identifier (Section C.1.1) and repeated three instruments from the pre-study survey: the Perceived Stress Scale (Section C.1.3), the Flourishing Scale (Section C.1.4), and the HEMA-R (Section C.1.5). The HEMA-R items were

unchanged, but the stem was reworded in the post-study instrument to “When participating in stress-relief activities, how strongly do the following intentions guide you?” Three additional instruments were introduced in the post-study survey to assess the experience of play, usability heuristics, and eudaimonic outcomes specific to engagement with *The Archive*.

C.2.1 Game Experience (GAME-X)

Response anchors for all GAME-X items: 1 = Strongly disagree, 2 = Somewhat disagree, 3 = Neither agree nor disagree, 4 = Somewhat agree, 5 = Strongly agree.

Absorption

Stem: “Rate your agreement with the following items in relation to how absorbed the game made you feel.”

1. While playing, I lost track of time.
2. While playing, I forgot my immediate surroundings.
3. Playing the game “got me away from it all.”

Agency

Stem: “Rate your agreement with the following items in relation to how much agency (control) the game provided you.”

1. While playing, I felt in charge.
2. While playing, I felt autonomous.
3. While playing, I felt confident.

C.2.2 Game Usability Heuristics (PLAY)

Stem: “Please rate your agreement with the following statements.” Response anchors: 1 = Strongly disagree, 2 = Somewhat disagree, 3 = Neither agree nor disagree, 4 = Somewhat agree, 5 = Strongly agree.

1. The activities did not feel repetitive.
2. The experience sustained my interest across the week.
3. Challenging moments felt manageable rather than overwhelming.
4. The pacing of the story and activities felt balanced.
5. *The Archive* story chapters responded to my choices in noticeable ways.
6. *The Archive* felt like a continuous world rather than disconnected episodes.
7. I understood what I was working toward in each chapter.
8. The story built toward a clear resolution.
9. I felt emotionally connected to the world of *The Archive*.
10. The story resonated with my experiences as a student.
11. I was able to begin using the app without confusion.
12. The experience did not feel mentally exhausting.

C.2.3 Eudaimonic Technology Experience Scale (ETES)

Stem: "Please rate your agreement with the following statements." Response anchors: 1 = Strongly disagree, 2 = Somewhat disagree, 3 = Neither agree nor disagree, 4 = Somewhat agree, 5 = Strongly agree.

1. I used this system to learn new things.
2. I used this system to pursue my aspirations.
3. This system stimulated me to pursue my goals.
4. When using this system, I felt a connection with my deepest feelings.
5. When using this system, I felt like I was in touch with who I truly am.
6. When using this system, I felt that my choices expressed my "true self."

C.3 Ecological Momentary Assessment (EMA)

The EMA was delivered within *The Archive* immediately after each completed story module, producing up to five responses per participant across the five-day deployment. A daily reminder notification was sent at a participant-selected time to prompt engagement with the day's module; the three EMA items followed the in-story check-in sequence and were presented only after module completion.

1. At this moment, how stressed do you feel? (*1 = Not at all stressed, 10 = Extremely stressed.*)
2. At this moment, how confident do you feel in your ability to cope with your current stressors? (*1 = Not at all confident, 10 = Extremely confident.*)
3. Was today's story worth your time? (*1 = Not at all, 2 = Slightly, 3 = Somewhat, 4 = Mostly, 5 = Completely.*)

C.4 Exit Interview Protocol

The exit interview was conducted during the in-person Day 6 session, immediately following completion of the post-study survey. Sessions were semi-structured and approximately 45 minutes in duration. The researcher asked each planned question in order and followed up with non-leading probing and clarification questions as needed to explore perceptions of comfort, support, narrative resolution, and overall stress reflection. Questions were organized into three blocks aligned to the research questions. Sessions were audio-recorded with participant consent and transcribed for analysis as described in Chapter 3.

C.4.1 Stress Experience

Aligned to RQ1.

1. During this week, did using *The Archive* change how you thought about or responded to stress? In what ways?

2. Did any particular story moment stand out as meaningful or difficult? Give us an example.
3. Did you feel more aware of your stress patterns after using the app? Tell us examples.

C.4.2 Game-Based Design Elements

Aligned to RQ2.

1. How did the branching choices affect your experience?
2. Did you feel supported by the story, or at any point uncomfortable or overwhelmed? How? Tell us about it.
3. How did journaling within the story influence your reflection?
4. How did the minigame feel in relation to the stress narrative?
5. What did the Lumen XP progression mean to you? Motivating? Neutral? Distracting?
6. Did the story endings feel resolved? Why or why not?
7. How did your experience with *The Archive* compare to other mental health apps you have used?

C.4.3 Reflection on Codesign

Aligned to RQ3.

1. Did the stress scenarios feel aligned with real student experiences? Why or why not?
2. Did the stories feel authentic? Why or why not?
3. Did anything feel unrealistic or disconnected from your lived stress? Why or why not?

4. What felt missing from the narratives?
5. If this were redesigned again, what would you change?
6. Did this feel different from apps like Headspace, Calm, or Finch? In what ways?
7. Did the narrative structure change how you engaged compared to more task-oriented apps? How? Tell us examples.
8. Did anything in *The Archive* feel like it replicated frustrations you've had with other wellness apps? How?